ARC RESEARCH · AI UPDATE

Learning on the job

Continual learning for LLMs: how frontier labs, neolabs and loop builders are betting on it



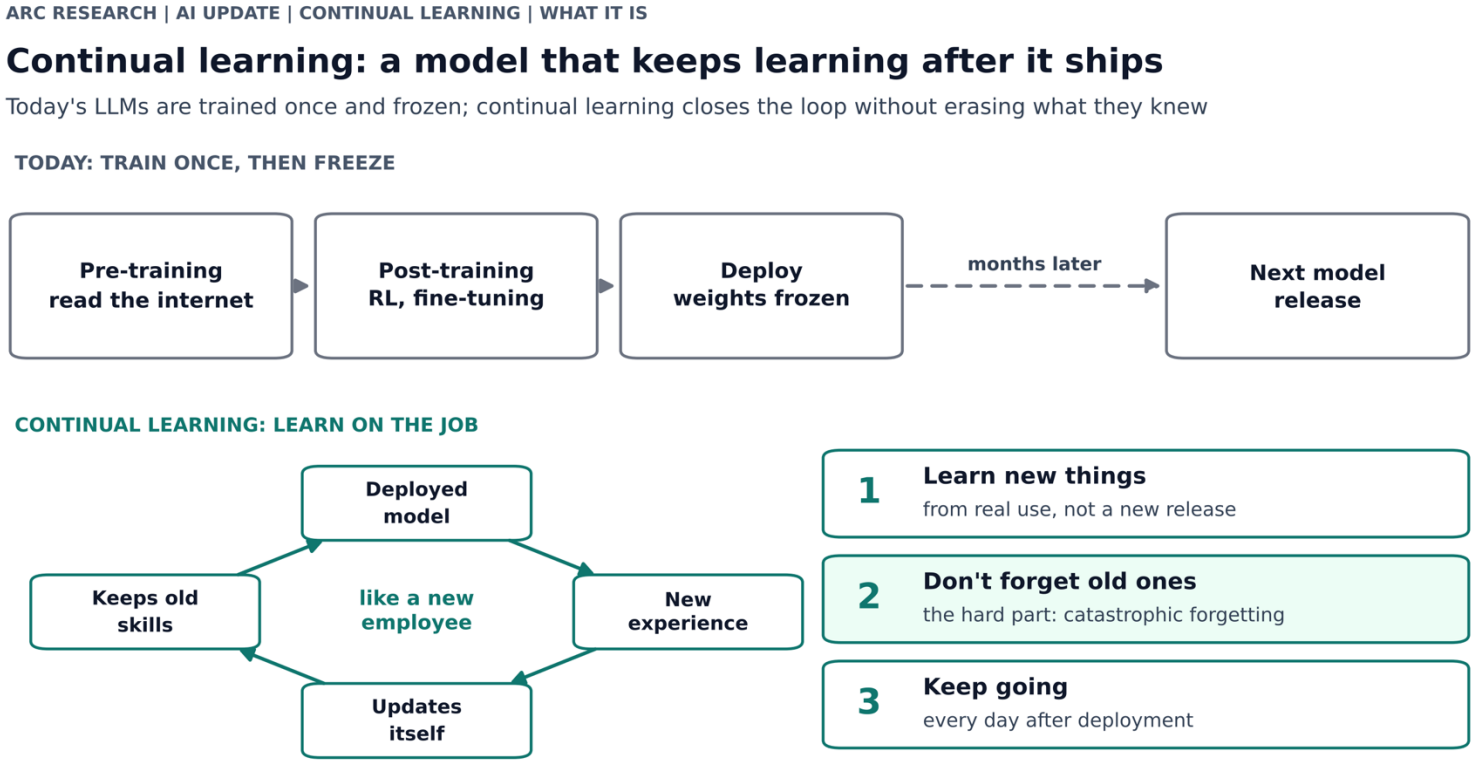
AGENDA

Six sections, from the problem to the bets

1	**Why it matters**	Frozen models, forgetting	Slides 5–7
2	**The toolbox**	Four approaches, the evidence	Slides 8–12
3	**Frontier labs**	Context vs. weights	Slides 13–16
4	**Neolabs & loop builders**	Lab by lab, loops, capital	Slides 17–31
5	**Recursive self-improvement**	Why AI that improves AI needs CL	Slides 32–38
6	**My opinion**	Context, my view, the evidence	Slides 39–43

Also: what it is and the map (slides 3–4), Sources (slide 44)

A model that keeps learning after it ships



SO WHAT

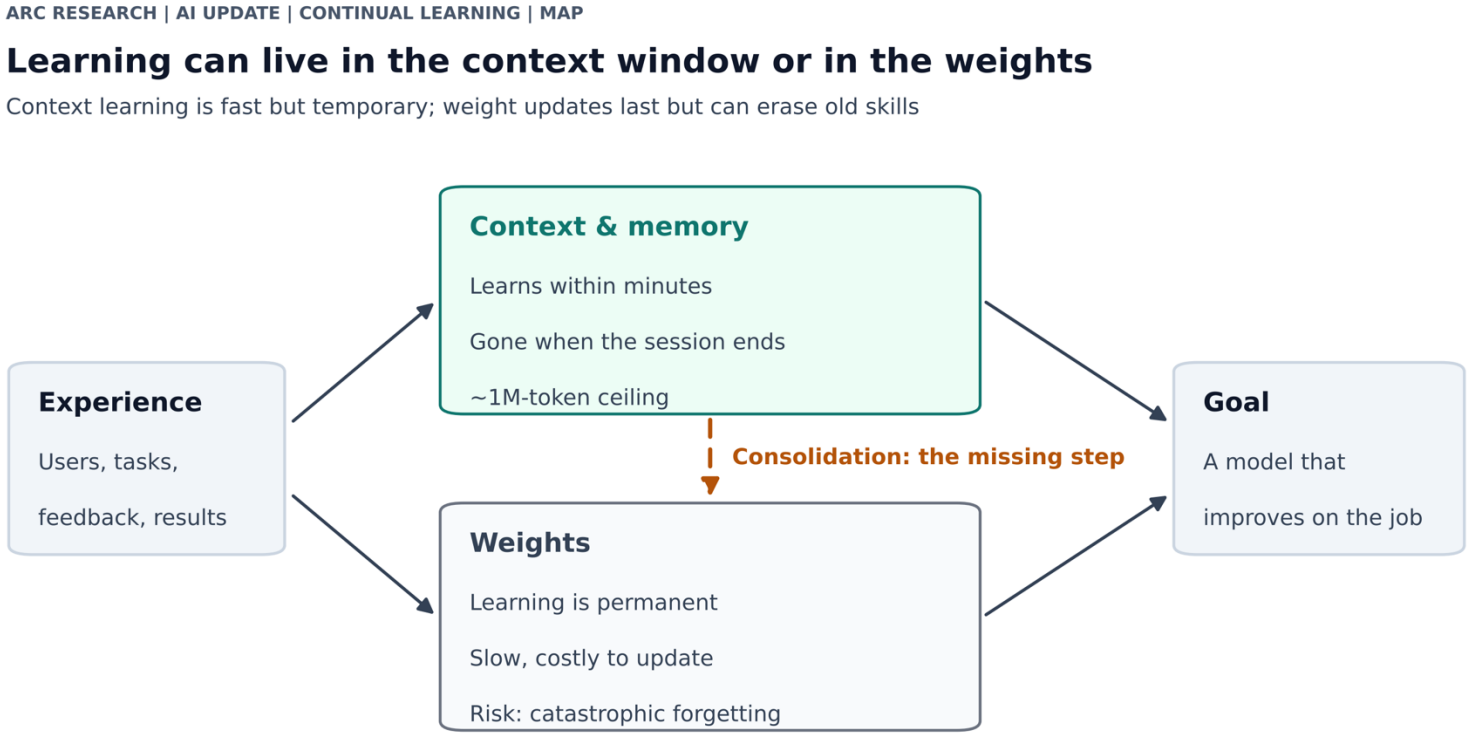
Today: train once, freeze, ship

Goal: learn on the job

Hard part: not forgetting

THE MAP

Two places learning can live



SO WHAT

Context: fast, but resets

Weights: durable, but overwrite old skills

The race: move context into weights safely

SECTION 1 OF 6

Why it matters

Today's models stop learning the day training ends

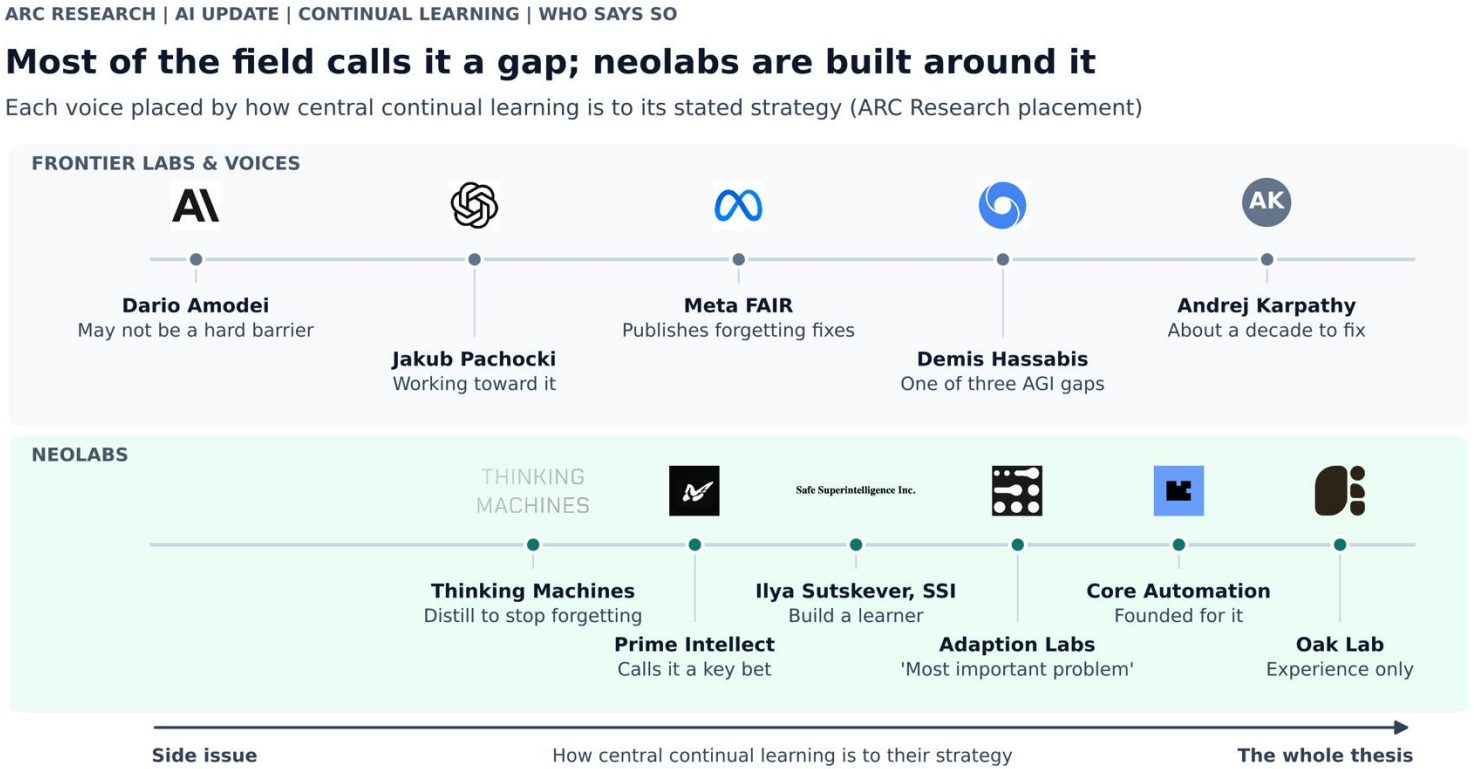
COMING UP

Frozen after training · 6

Updates erase old knowledge · 7

WHY IT MATTERS

Who calls it a gap, and who bets on it



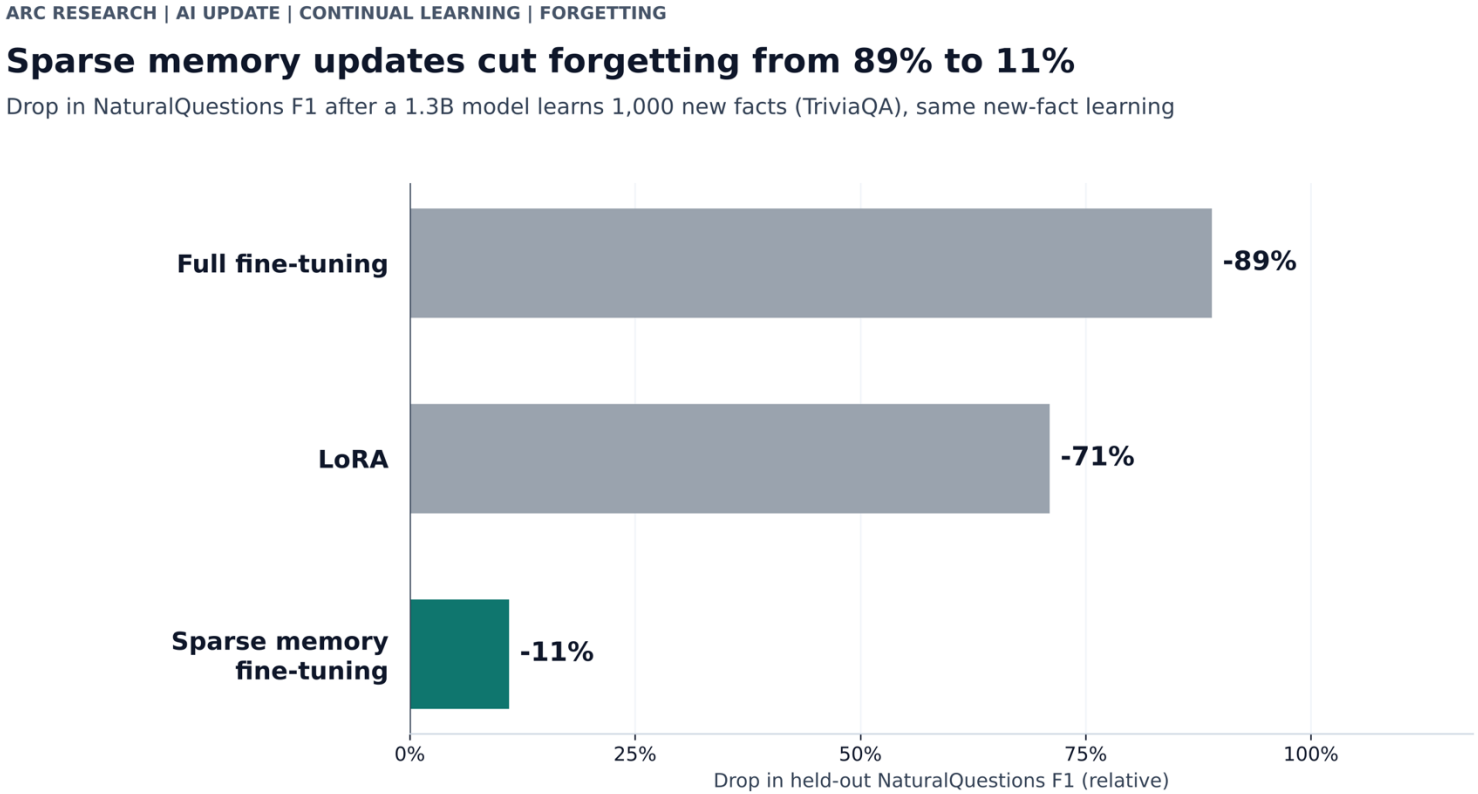
SO WHAT

Frontier labs: a gap, not yet the core

Neolabs: built around it

Lone skeptic: Anthropic's Amodei

Naive updates erase old knowledge



SO WHAT

Full fine-tune: 89% of old recall lost

LoRA alone barely helps: 71%

Touch fewer weights, forget less: 11%

SECTION 2 OF 6

The toolbox

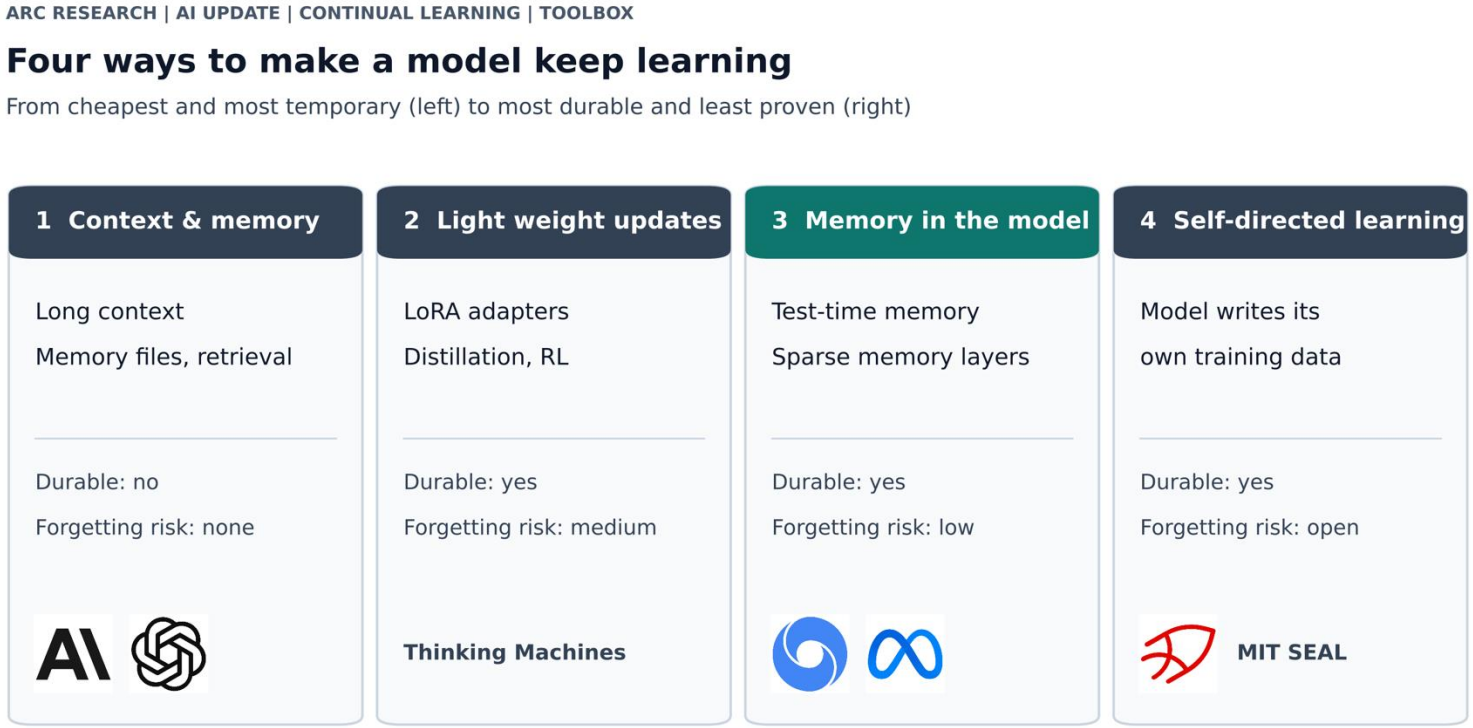
Four ways to keep learning, and what the papers show

COMING UP

Four approaches · 9
Distillation fixes forgetting · 10
Memory architectures · 11
Self-written training data · 12

THE TOOLBOX

Four ways to make a model keep learning



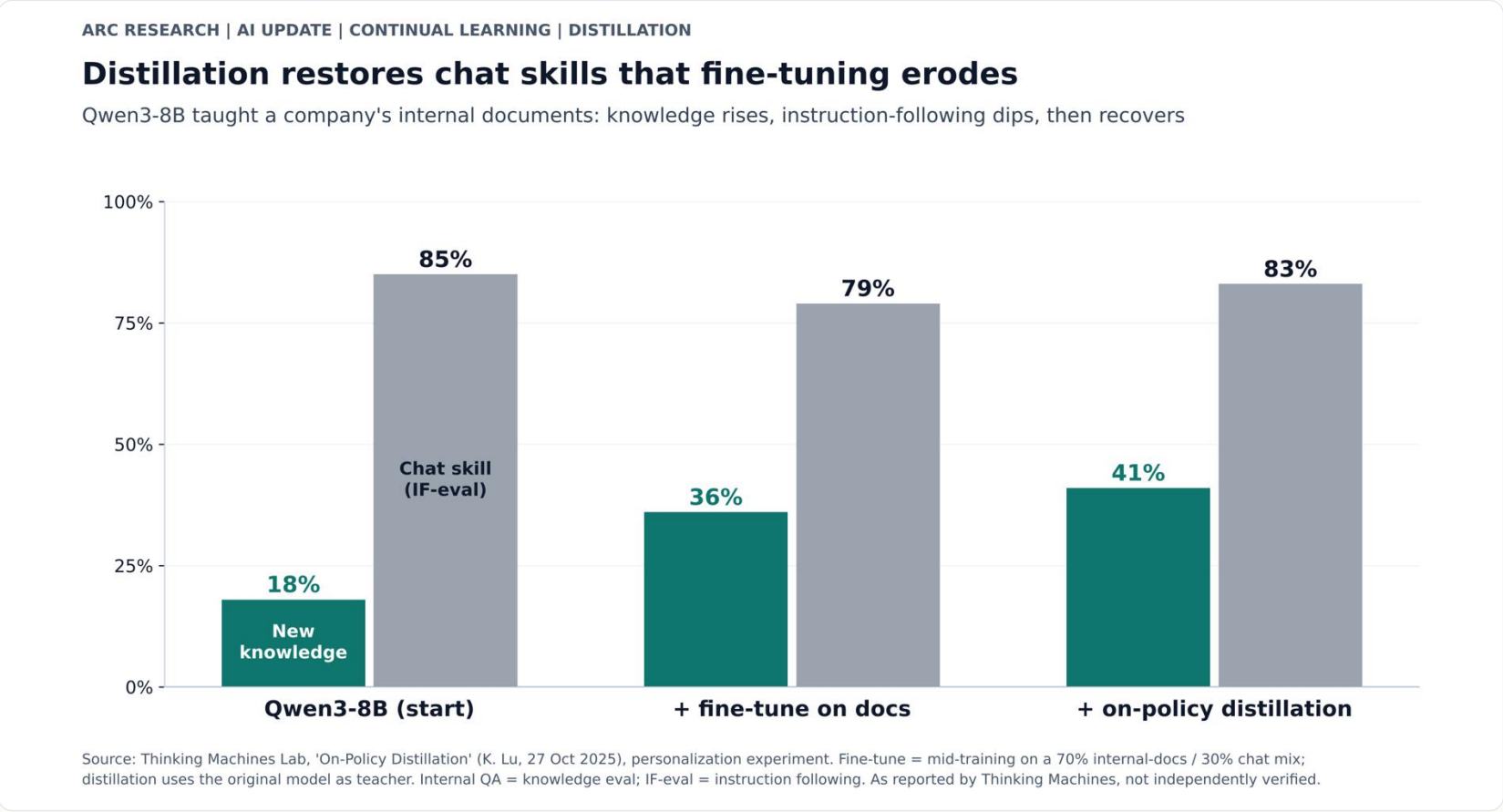
SO WHAT

Shipping today: context and memory

Research frontier: weights that update safely

Next three slides: one result each

Distillation repairs what fine-tuning breaks



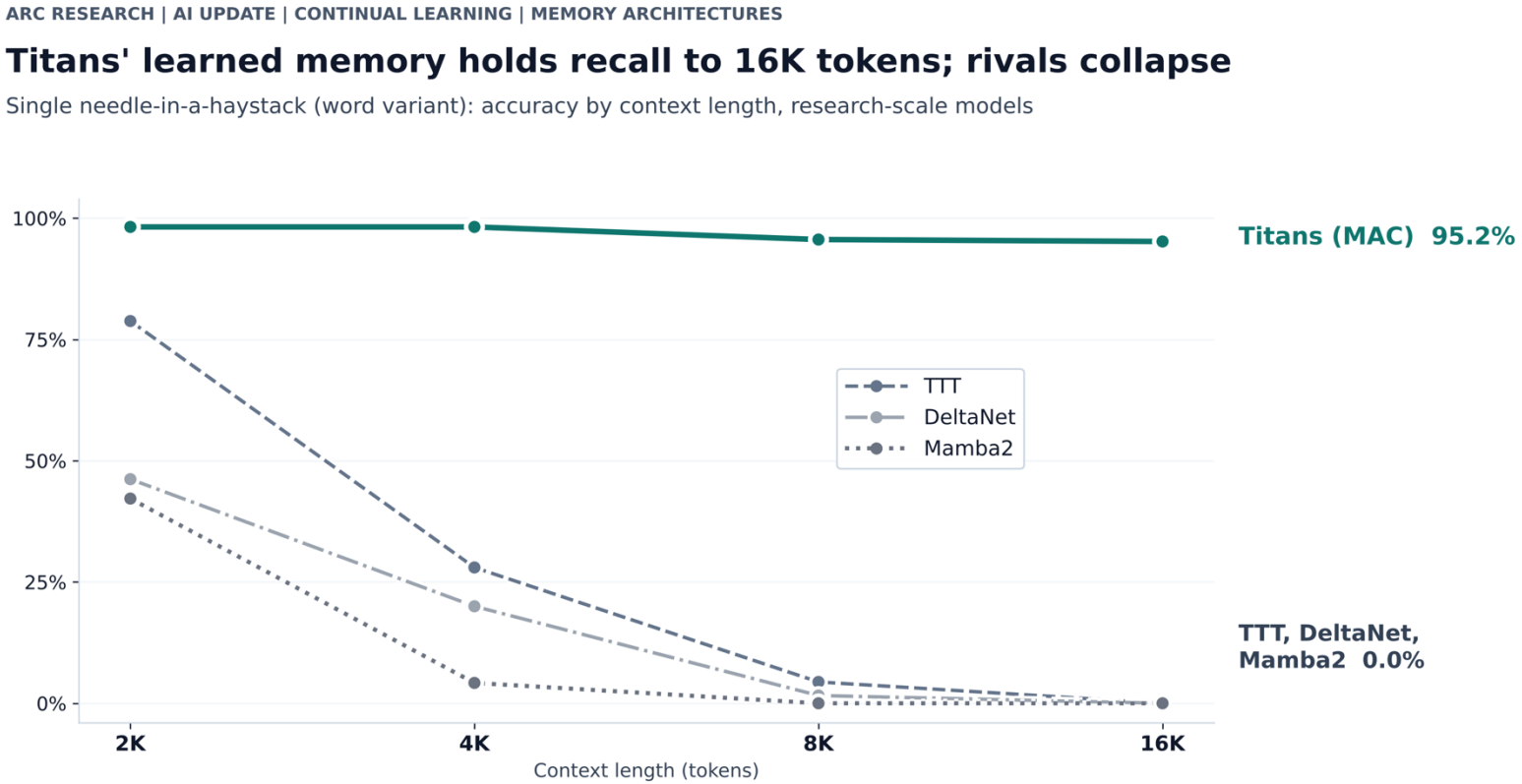
SO WHAT

New knowledge: 18% → 41%

Chat skill dips 85% → 79%, back to 83%

Recipe: train, then distill, repeat

A learned memory holds recall at length



SO WHAT

Titans: 95.2% recall at 16K

Rival recurrent models: 0%

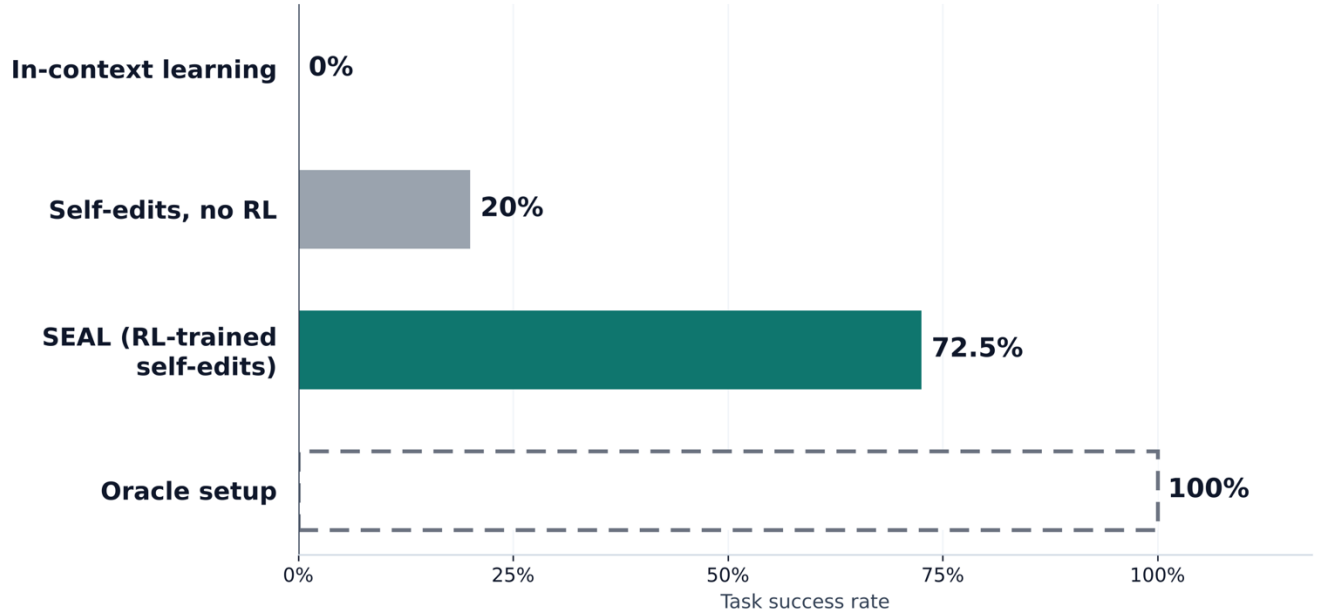
Research scale; not in Gemini yet

Models can write their own training data

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A model that writes its own training data solves 72.5% of ARC tasks

Success rate on held-out few-shot ARC puzzles, Llama-3.2-1B-Instruct



Source: Zweiger, Pari, Guo, Akyurek, Kim, Agrawal (MIT), 'Self-Adapting Language Models' (SEAL), arXiv 2506.10943, Table 1; NeurIPS 2025.
Oracle TTT = hand-designed best training setup (upper bound, hollow). Authors note SEAL still forgets earlier tasks as edits accumulate. As reported by the authors.

SO WHAT

Prompting alone: 0%

Self-chosen updates: 72.5%

Still forgets as edits pile up

SECTION 3 OF 6

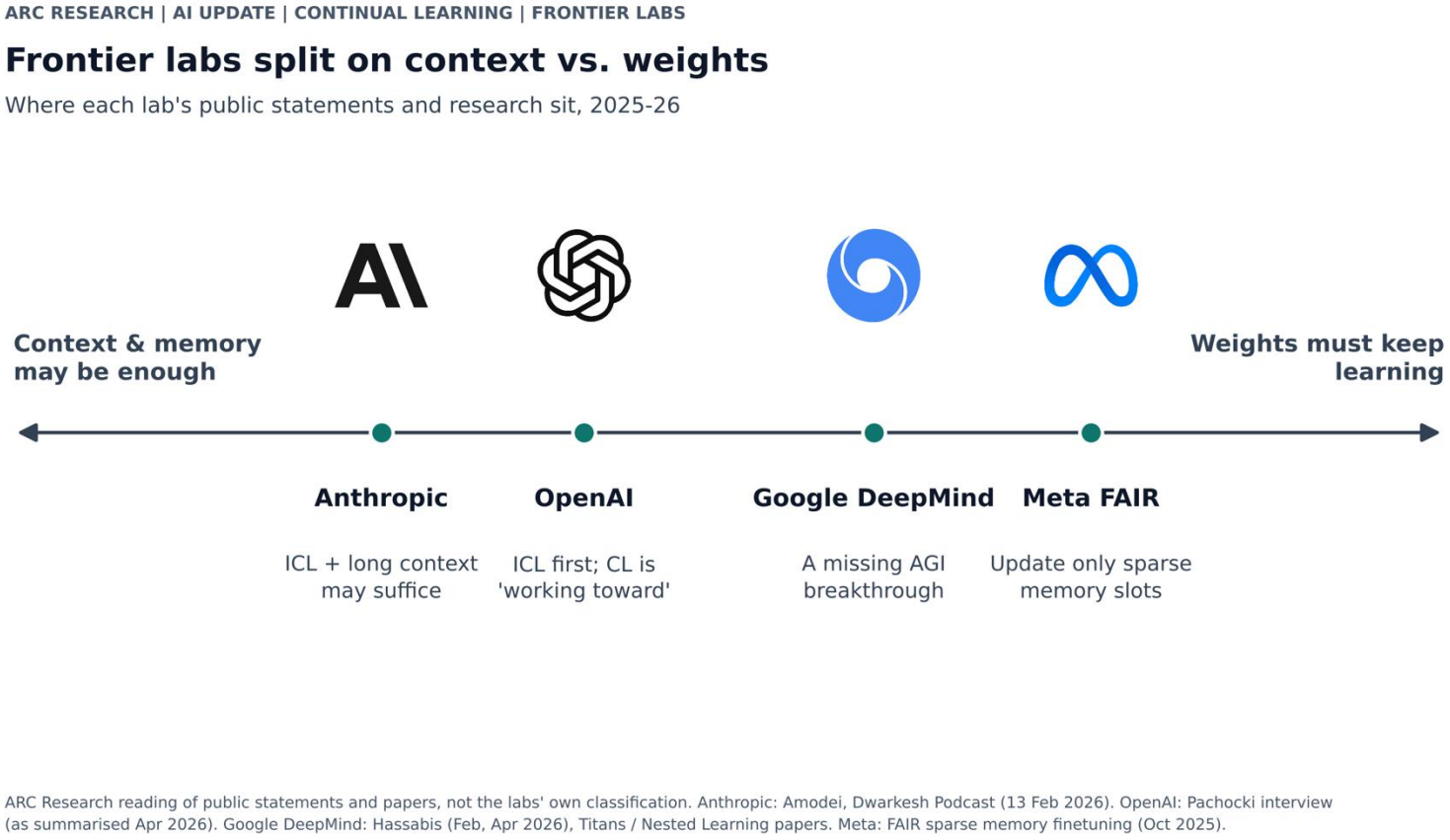
Frontier labs

Context may be enough vs. weights must keep learning

COMING UP

Where each lab sits · 14
Anthropic & OpenAI · 15
Google DeepMind & Meta · 16

The labs split on context vs. weights



SO WHAT

All four are working on it

Split: how much context can carry

No frontier model updates weights live

Anthropic and OpenAI: context may carry it

Anthropic

- Maybe not a hard barrier
- 1M tokens ≈ days of human learning
- "Good chance" solved in 1–2 years
- Product: memory, free to all

OpenAI

- CL: what it is "working toward"
- In-context beats RL on efficiency
- Evals in context vs. custom models
- Product: memory across all chats

DeepMind and Meta: build memory into the model

Google DeepMind

- CL: one of three AGI gaps
- Titans: memory learned at test time
- Nested Learning / HOPE
- Gemini leads: slower than hoped

Meta

- FAIR: sparse memory fine-tuning
- Forgetting: 89% → 11%
- Research, not product
- MSL: no public CL position

SECTION 4 OF 6

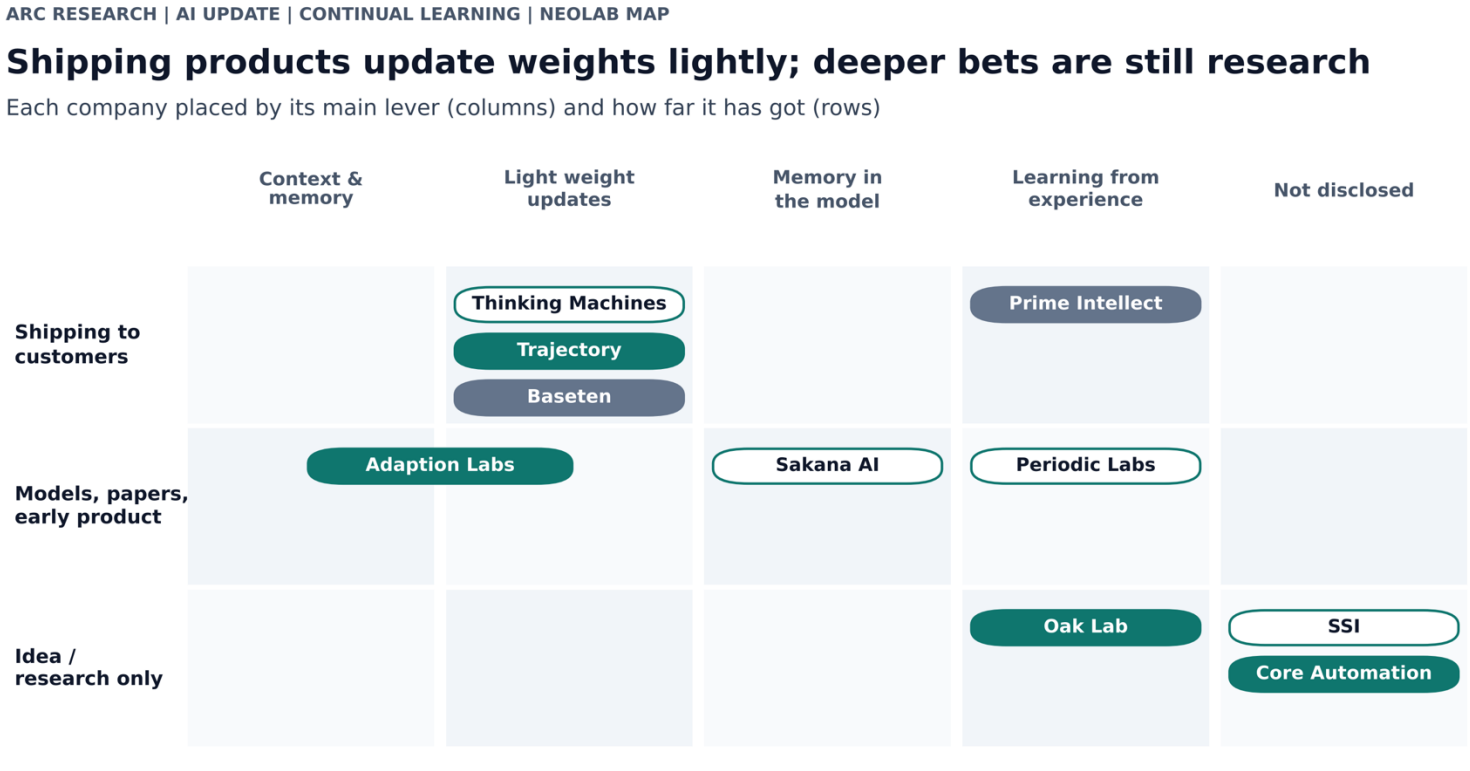
Neolabs & loop builders

Who is betting the company on learning from experience

COMING UP

The map: lever vs. stage · 18
Three learning loops · 30
The capital behind them · 31
Harness, model or both · 19
Lab by lab, ten companies · 20–29

Ten companies, mapped by lever and stage



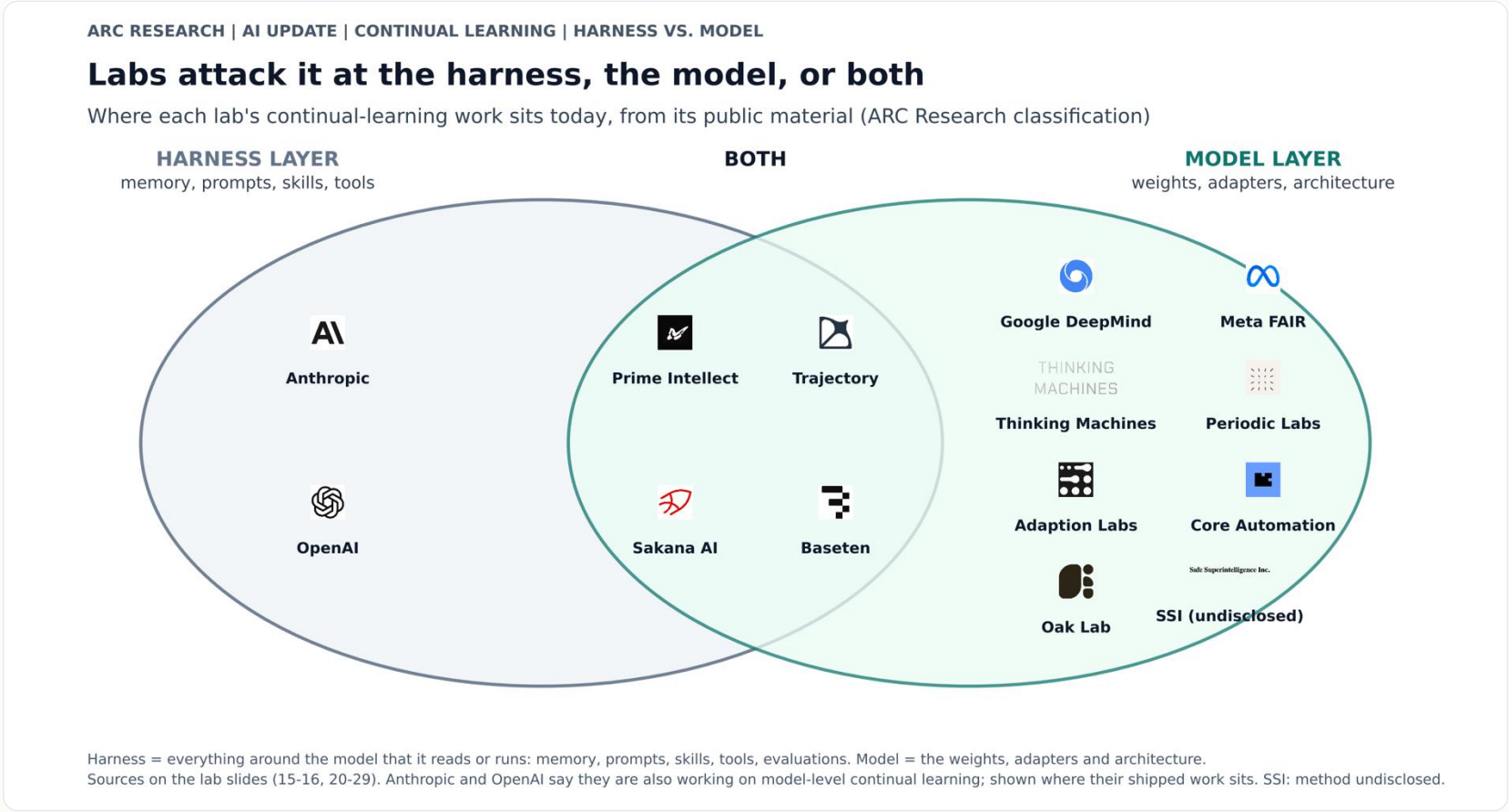
SO WHAT

Shipping today: light weight updates

Deep bets: still research

Lab by lab next: slides 20–29

Harness, model, or both



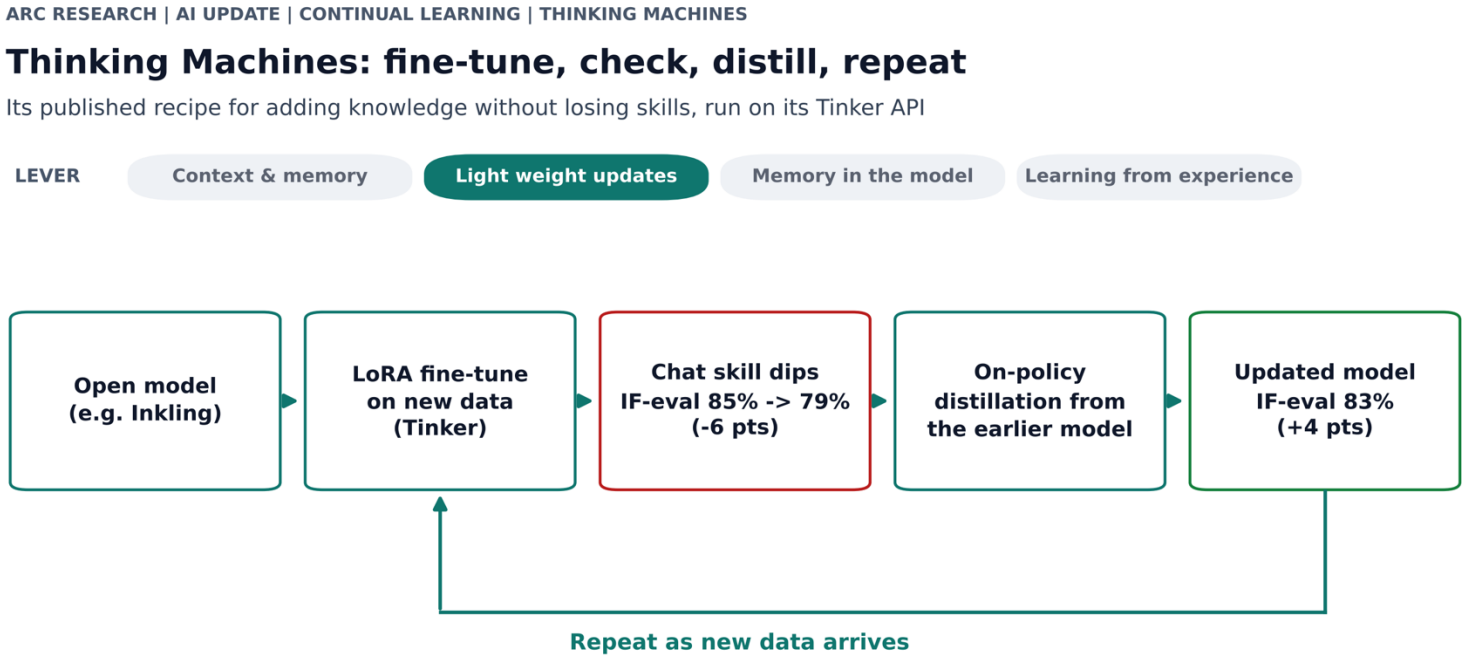
SO WHAT

Harness: Anthropic, OpenAI ship memory

Model: most neolabs bet on weights

Both: loop builders tune each

Keep updating the weights, cheaply



SO WHAT

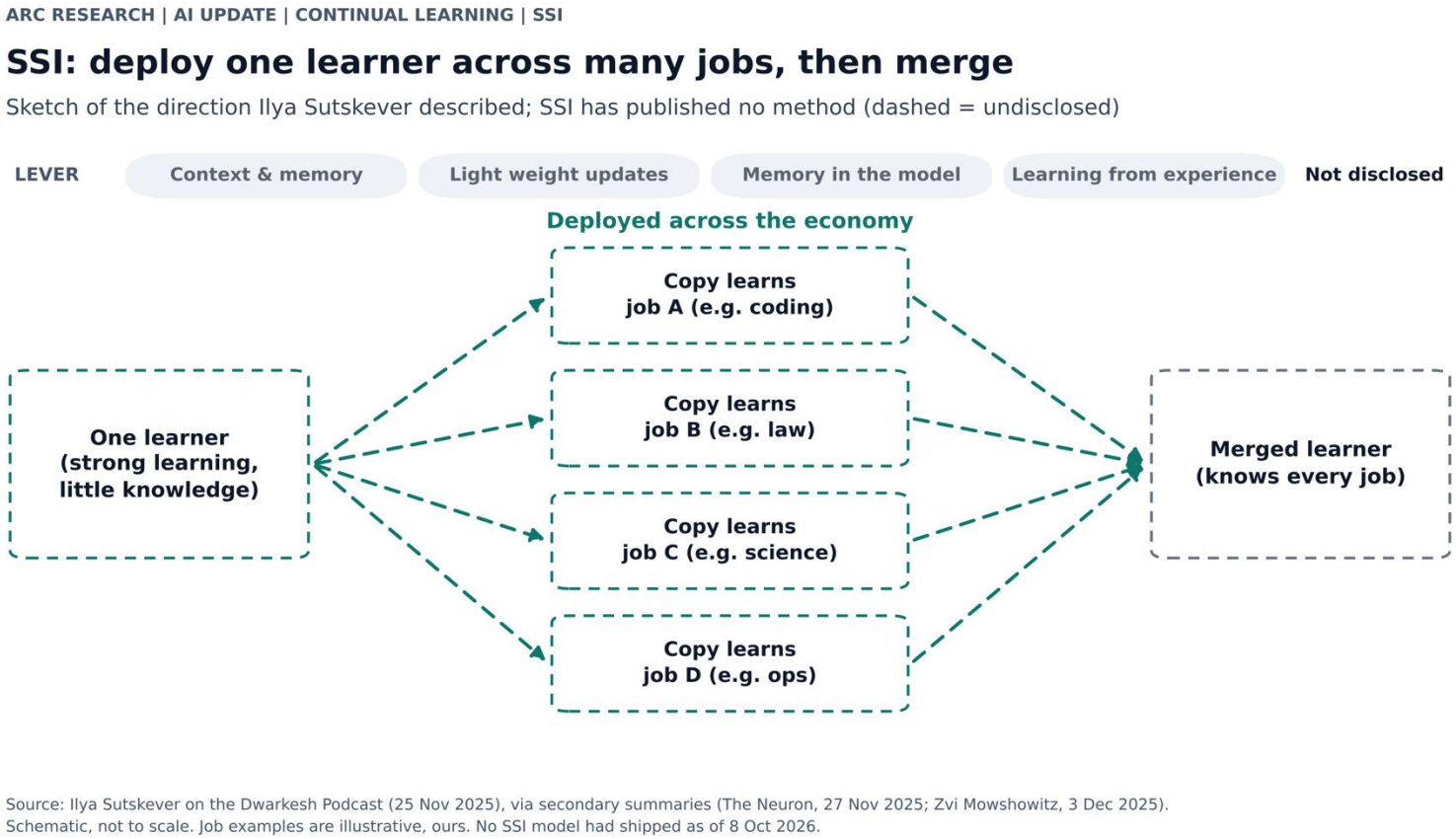
Fine-tune, then distill away the forgetting

Tinker sells the loop

$2B seed; $5-6B round in talks

Safe Superintelligence Inc.

Build a learner, then let it learn on the job



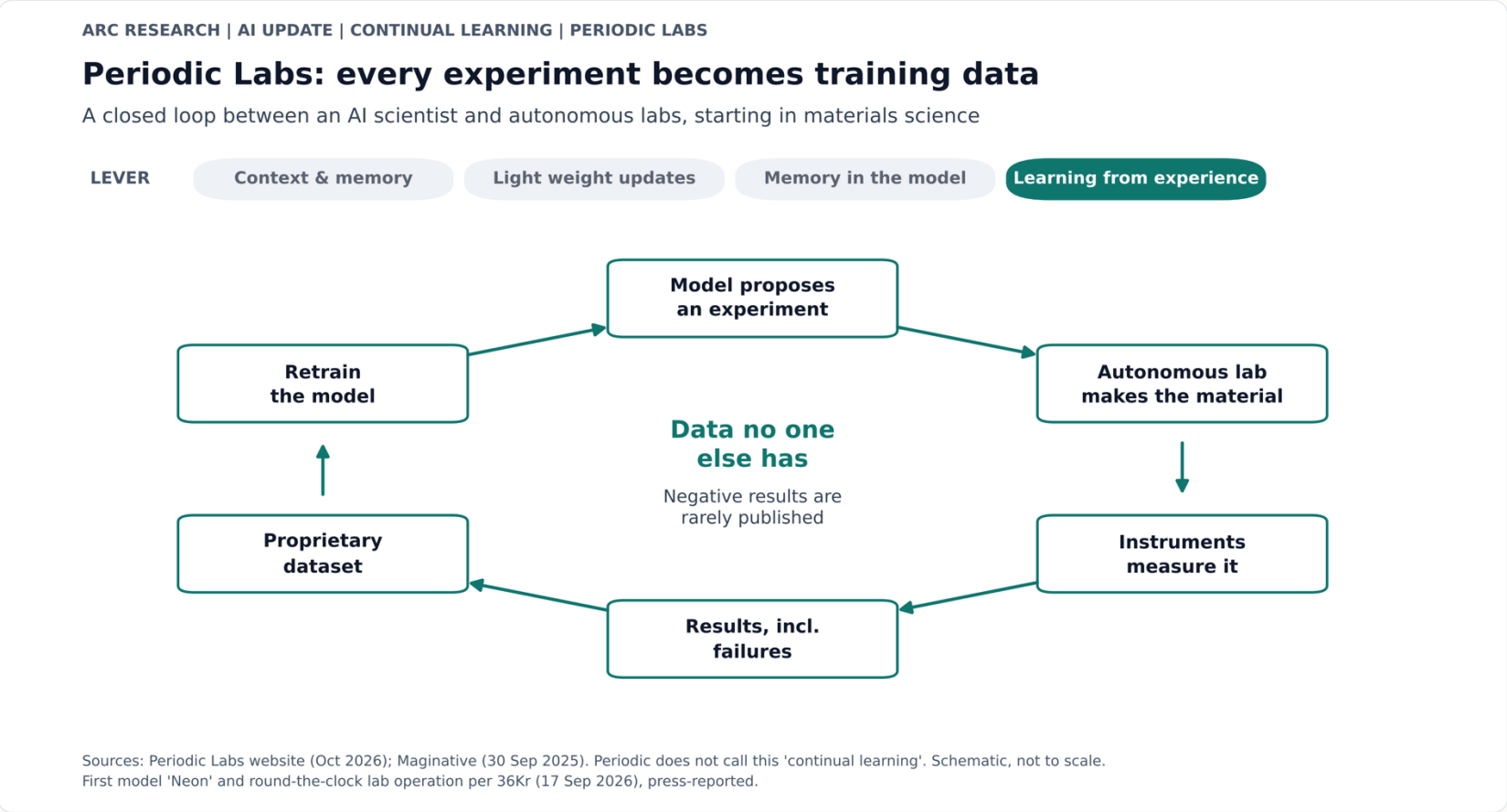
SO WHAT

Learner first, knowledge later

Method undisclosed

No model shipped yet

Learn from experiments no one else can run



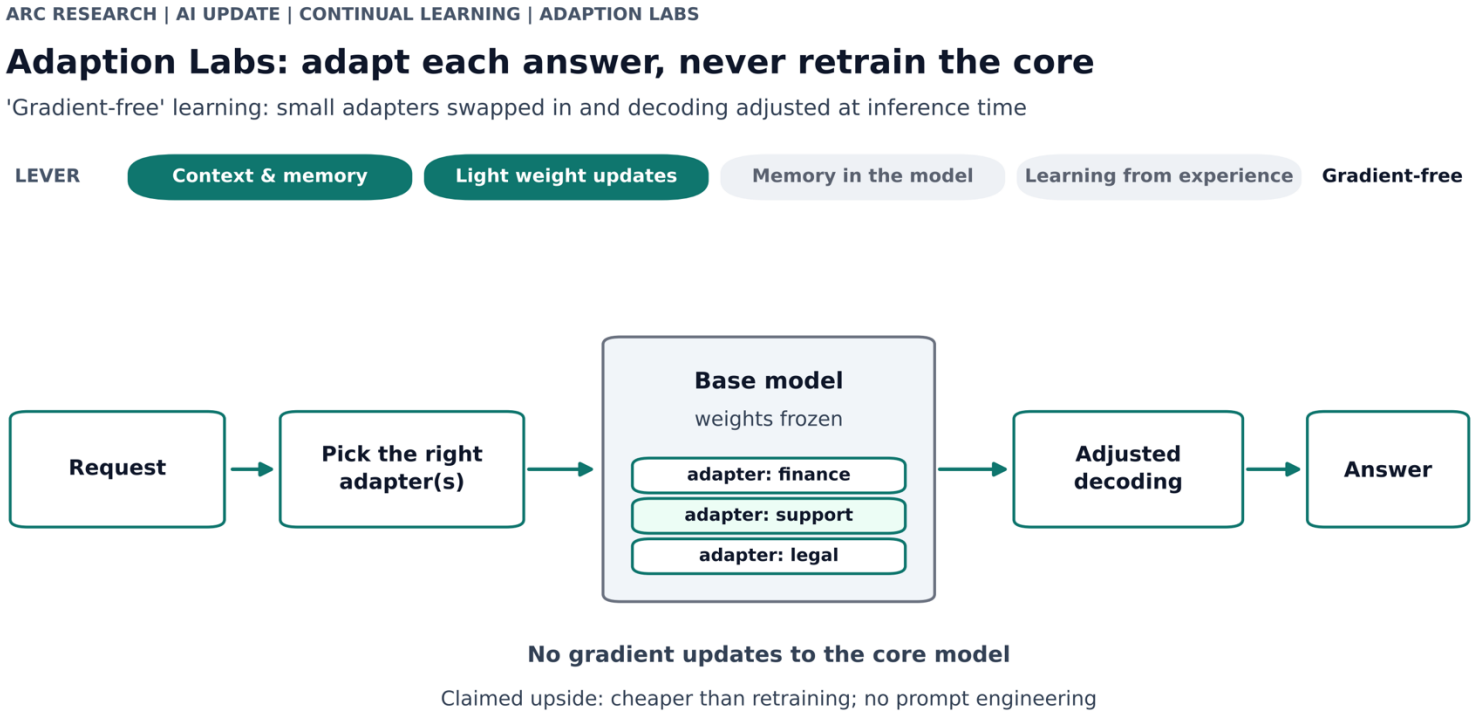
SO WHAT

Moat: lab data no one else has

Failures count as training data

First model, Neon (Sep 2026)

Adapt at inference; leave the weights alone



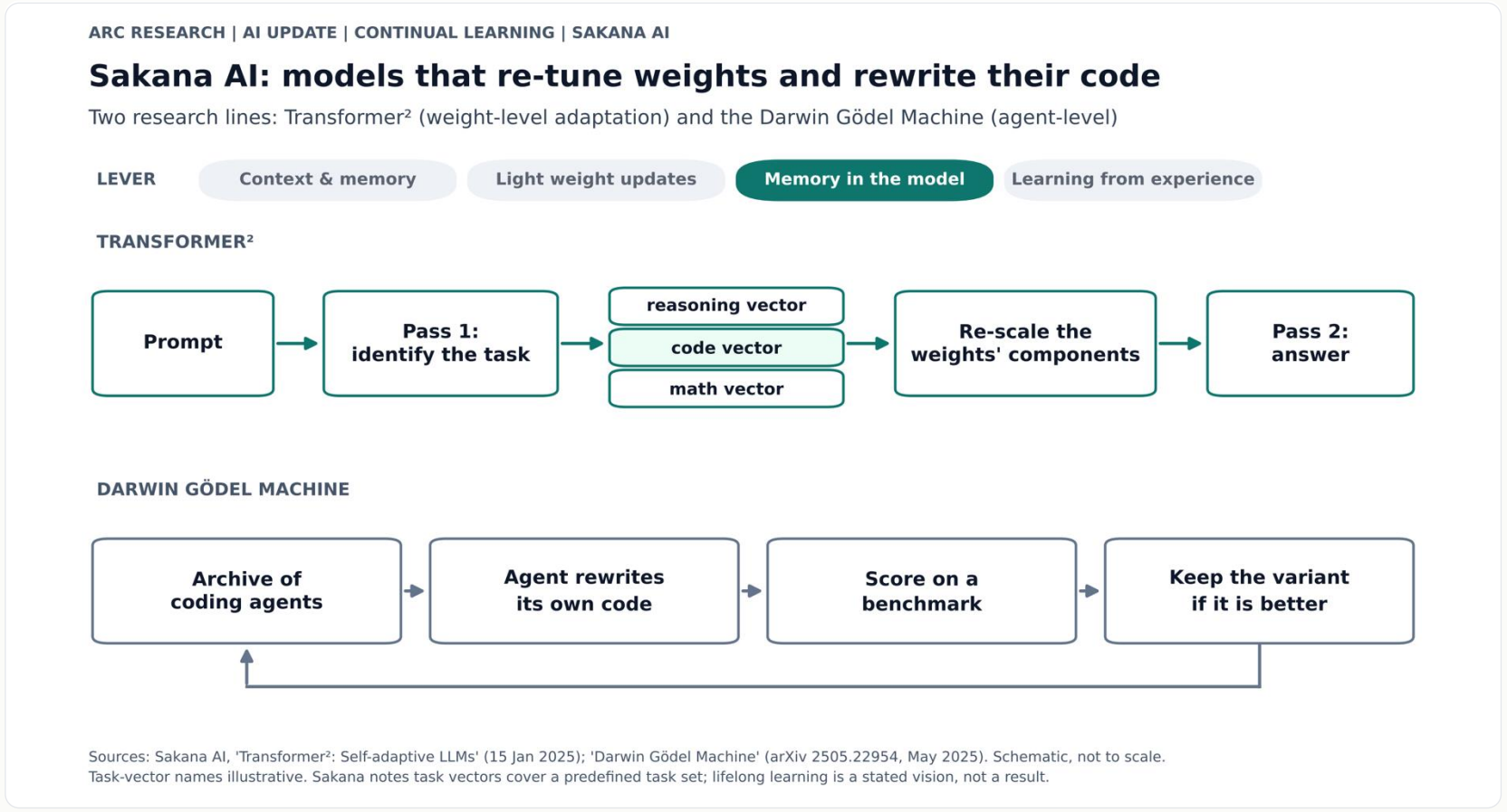
SO WHAT

Core weights never retrained

Adapts per request

$50M seed (Feb 2026)

Weights that re-tune themselves per task



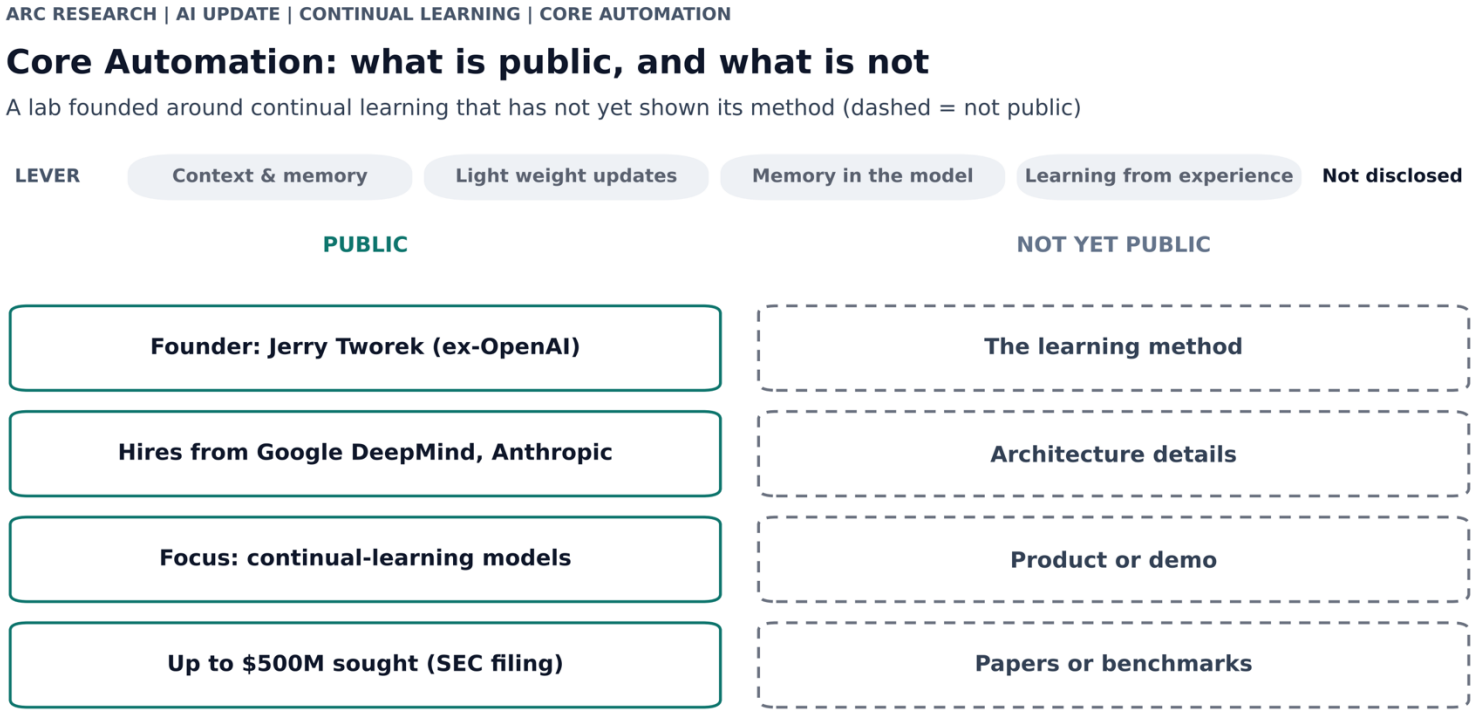
SO WHAT

Weights re-scale per task

Agents rewrite their own code

Not yet open-ended learning

A lab founded to solve continual learning



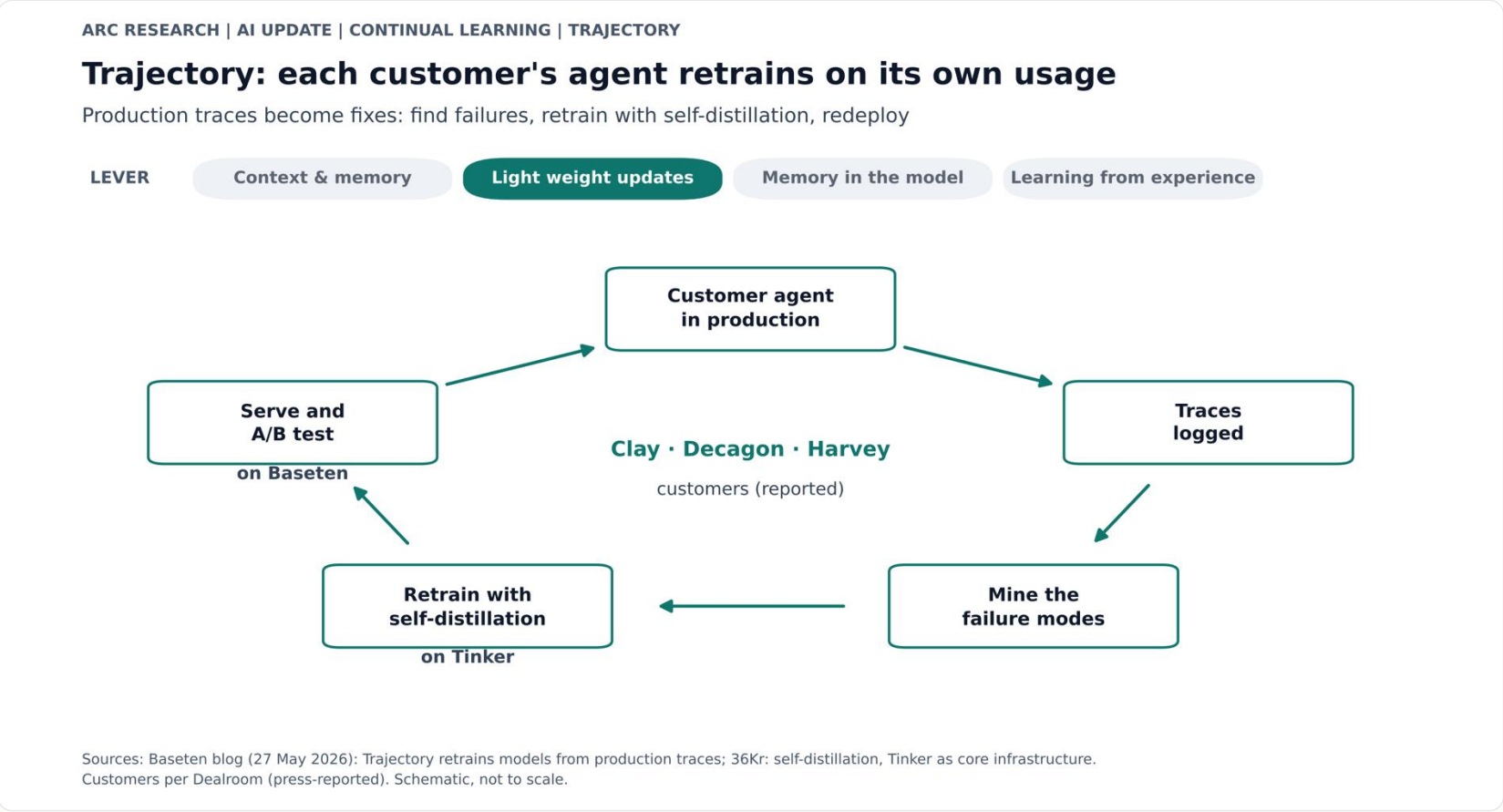
SO WHAT

Built purely for continual learning

Method still private

Up to $500M sought, not closed

Retrain each customer's agent on its own traces



SO WHAT

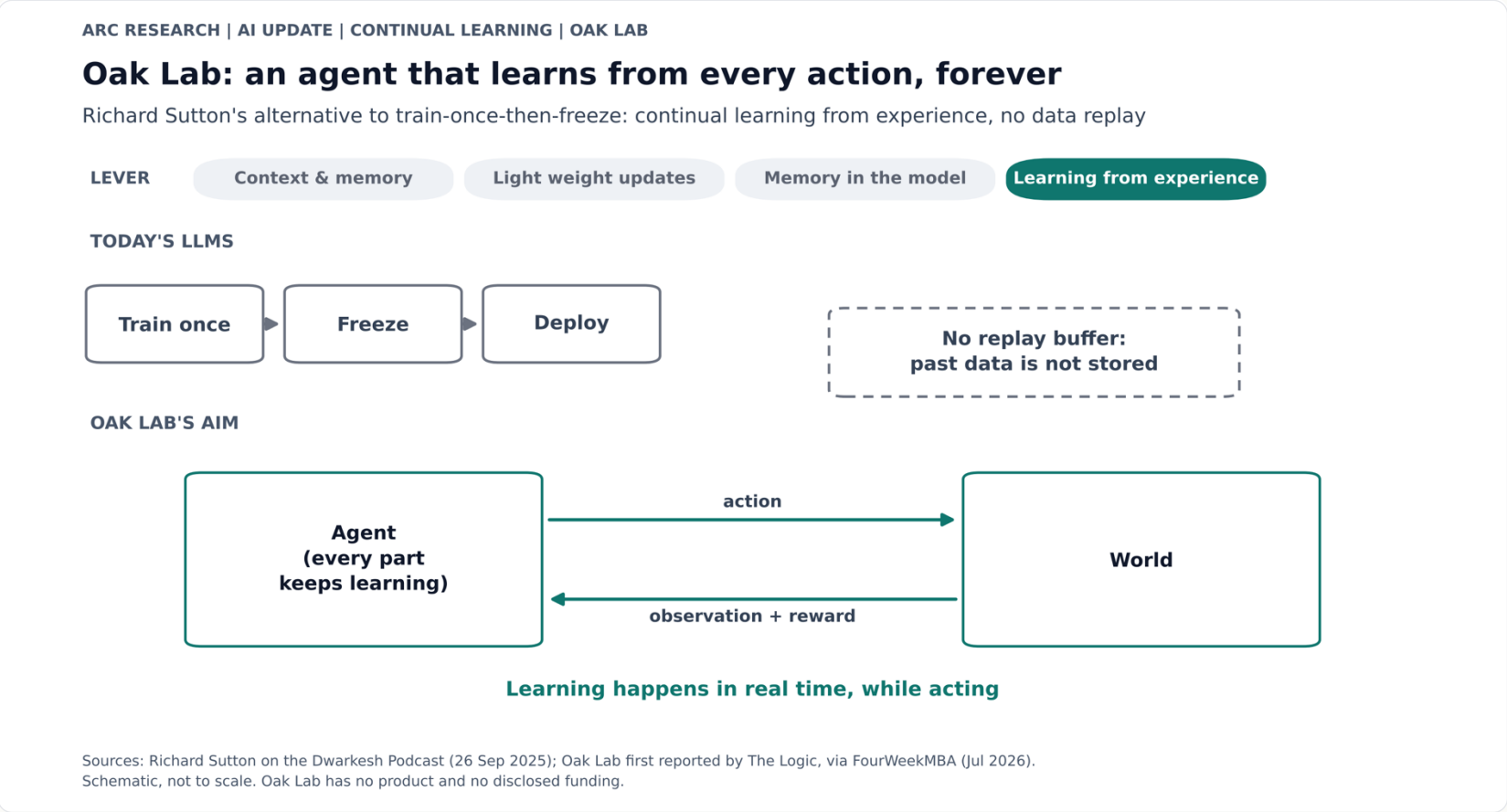
Learns from each customer's usage

Runs on Tinker and Baseten

$40M Series A (reported)

Oak Lab

Sutton's bet: learn only from experience



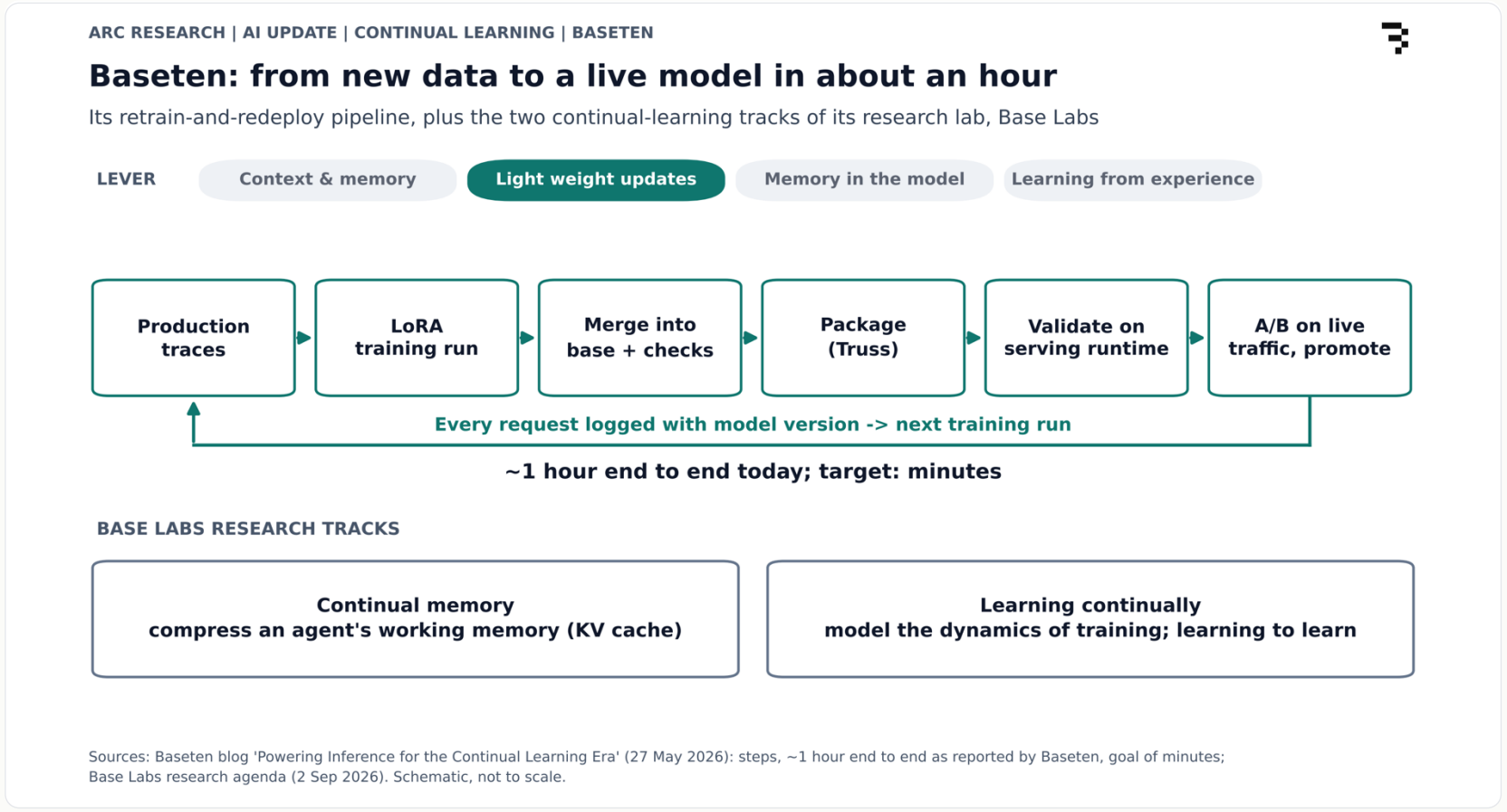
SO WHAT

Learning never stops

No stored data replayed

No product, no disclosed funding

Turn retrain-and-redeploy into an hourly loop



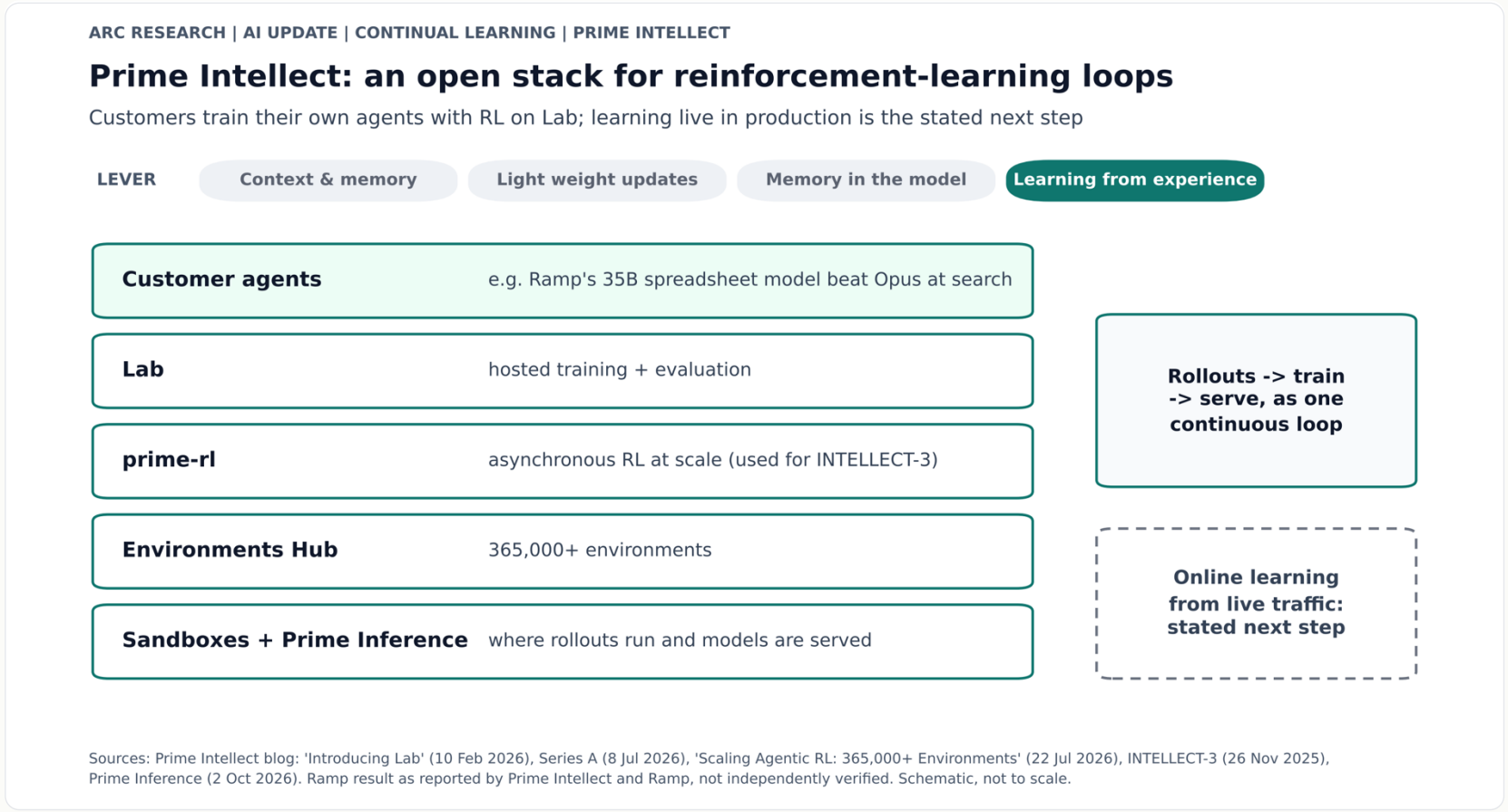
SO WHAT

Retrain to live in ~1 hour

A/B test before promotion

Base Labs: memory research

Training and inference as one RL loop



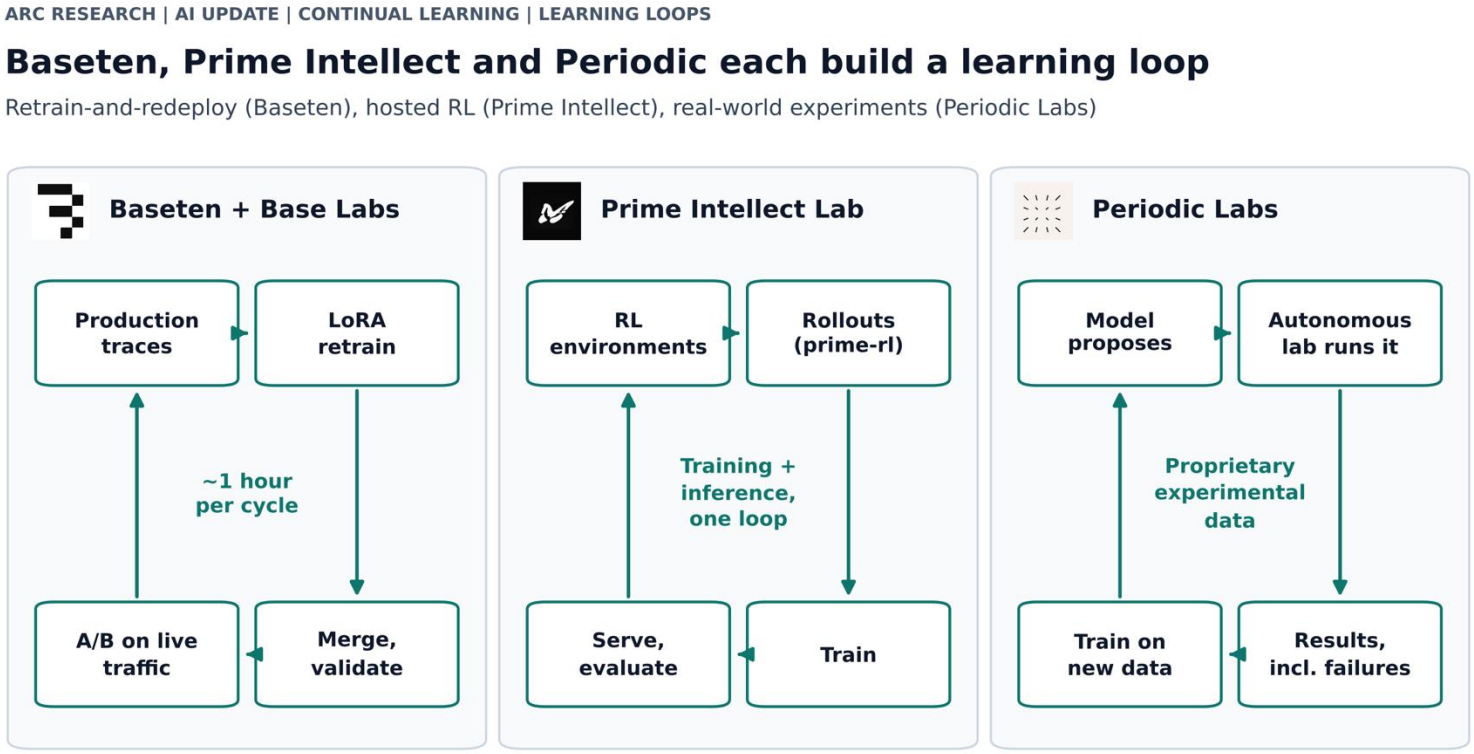
SO WHAT

Customers own their RL loop

Ramp's 35B beat Opus at one task

Live online learning: next step

Three ways to close the learning loop



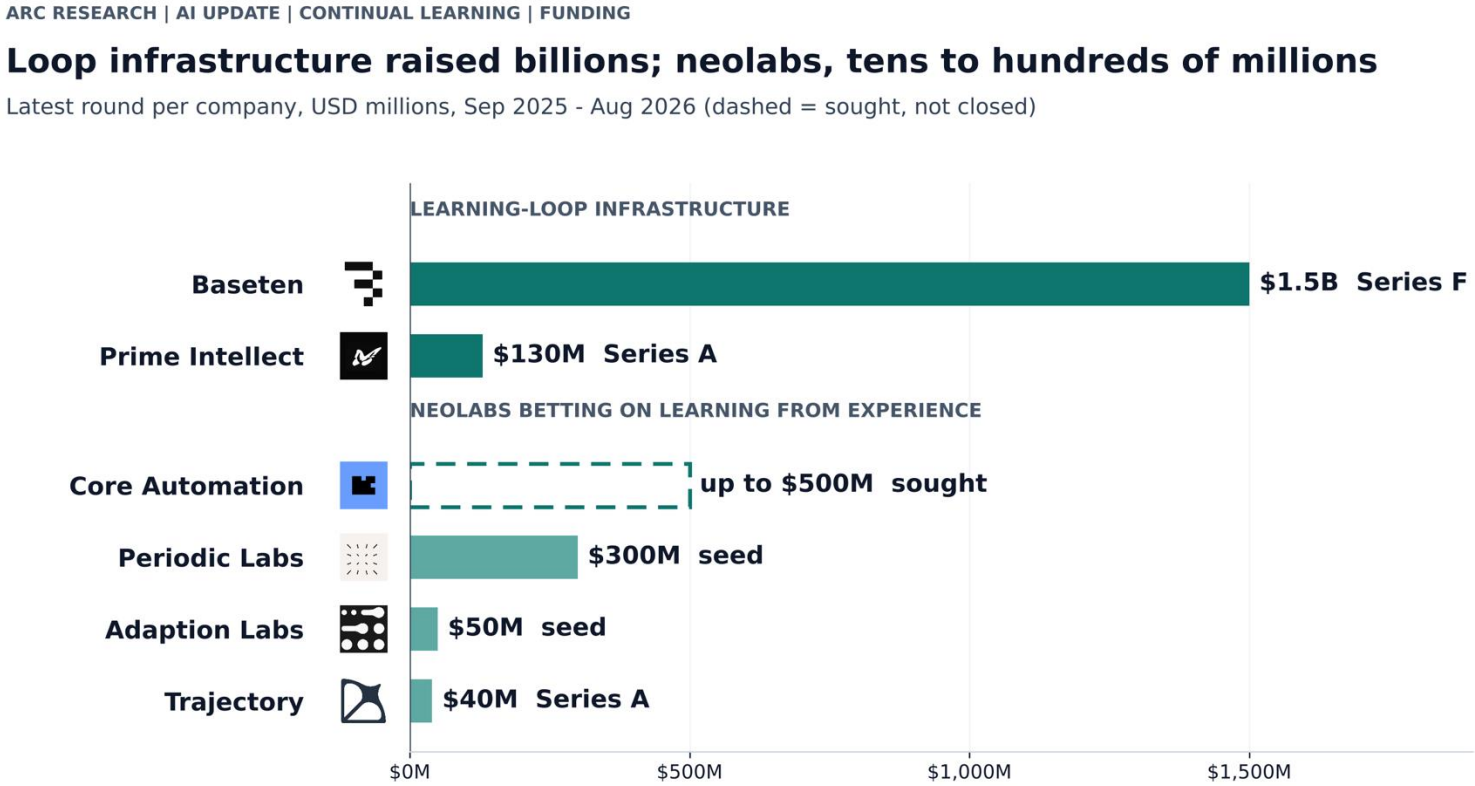
SO WHAT

Baseten: retrain, redeploy, ~1 hour

Prime Intellect: hosted RL, one loop

Periodic: data no one else has

The money sits with the loop builders



SO WHAT

Baseten: $1.5B, the largest round

Biggest CL-native ask: $500M, not closed

Tooling is funded before the science

SECTION 5 OF 6

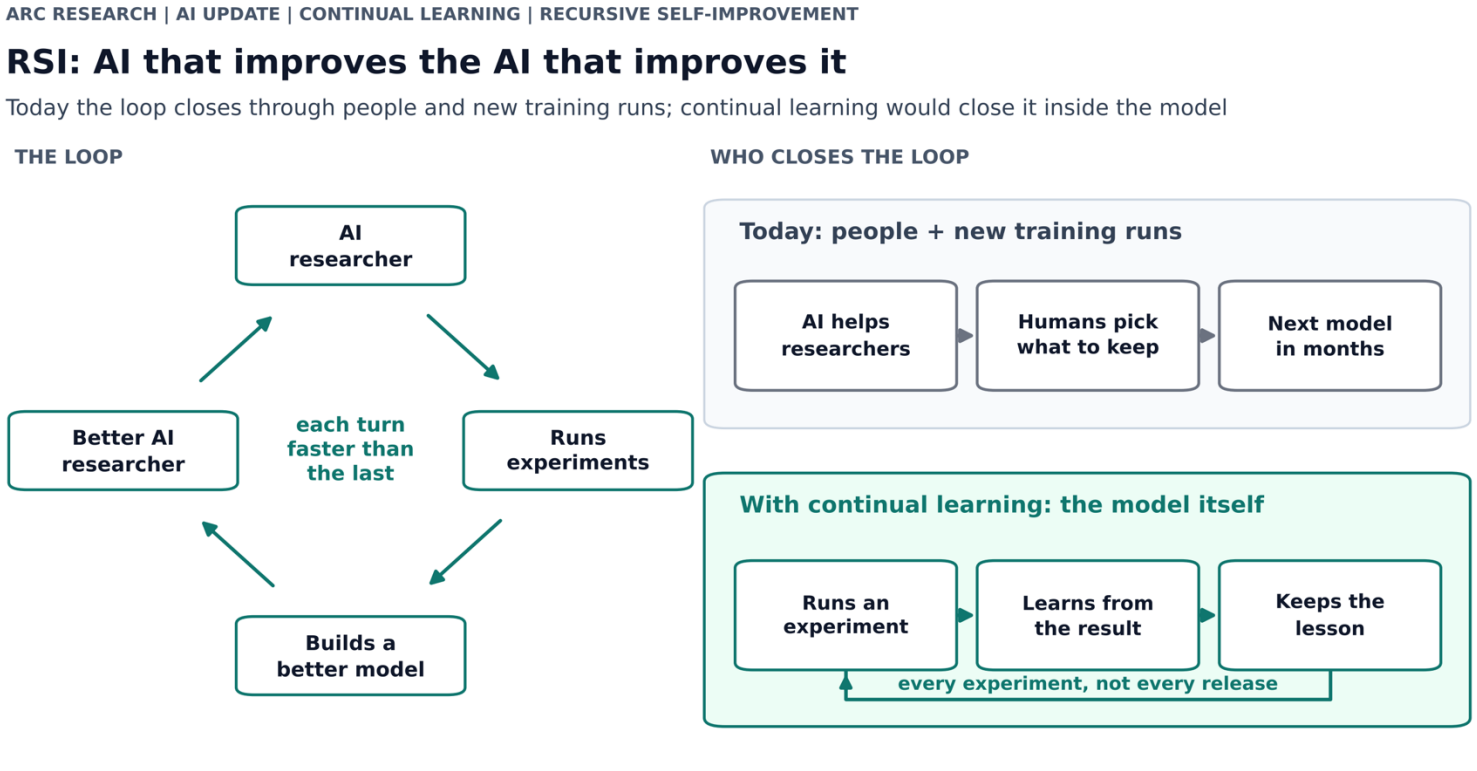
Recursive self-improvement

Why continual learning is the missing link

COMING UP

What RSI is · 33
How long AI works alone · 34
AI speeding up AI · 35
Why it needs CL · 36
Who says when · 37
Guardrails · 38

AI that improves the AI that improves it



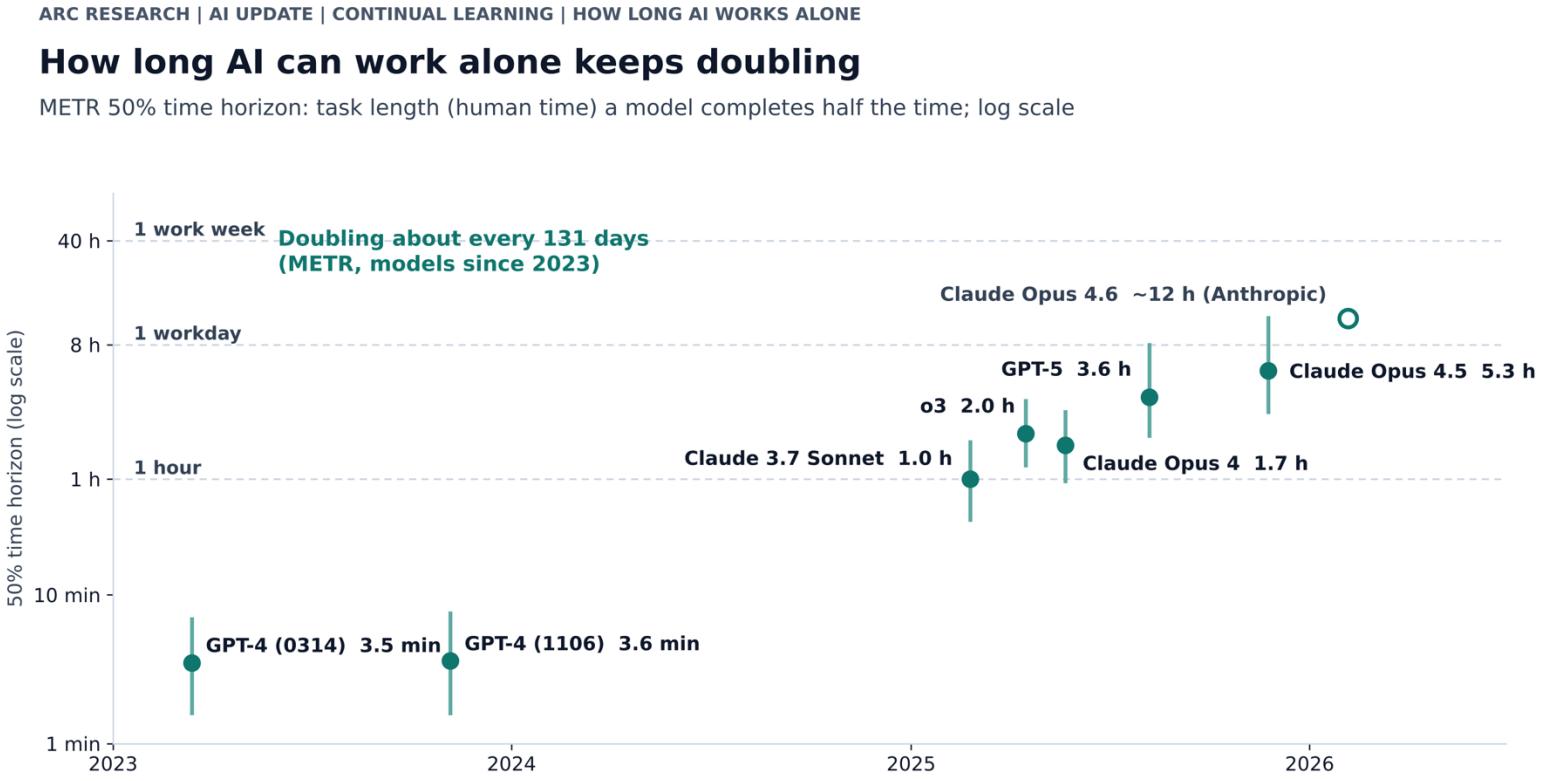
SO WHAT

RSI: AI builds better AI

Today: people close the loop

With CL: the model closes it

AI now works alone for hours, not yet weeks



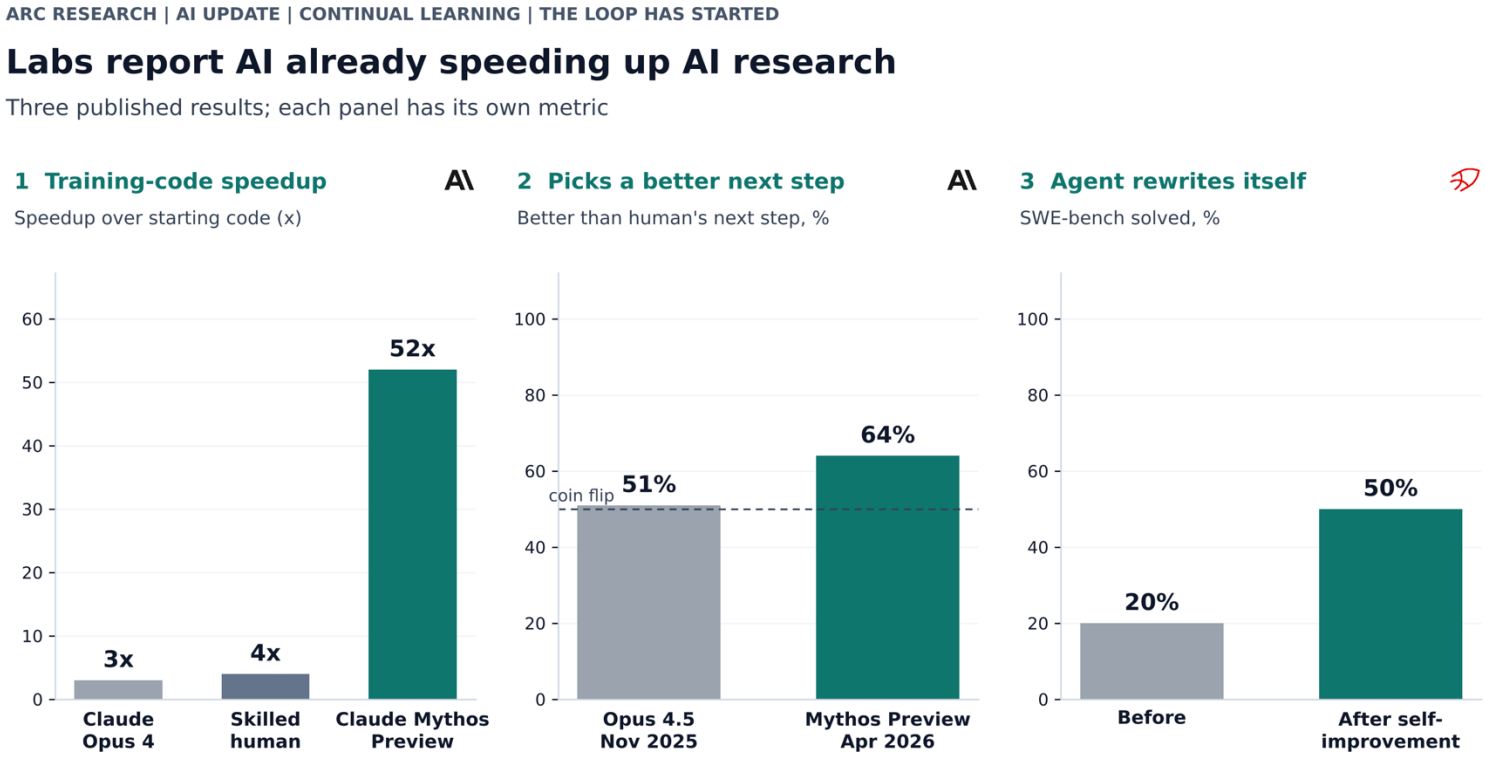
SO WHAT

2023: 3.5 min → late 2025: 5.3 h

Doubling about every 131 days

Research takes weeks: needs memory

Labs report AI already speeding up AI research



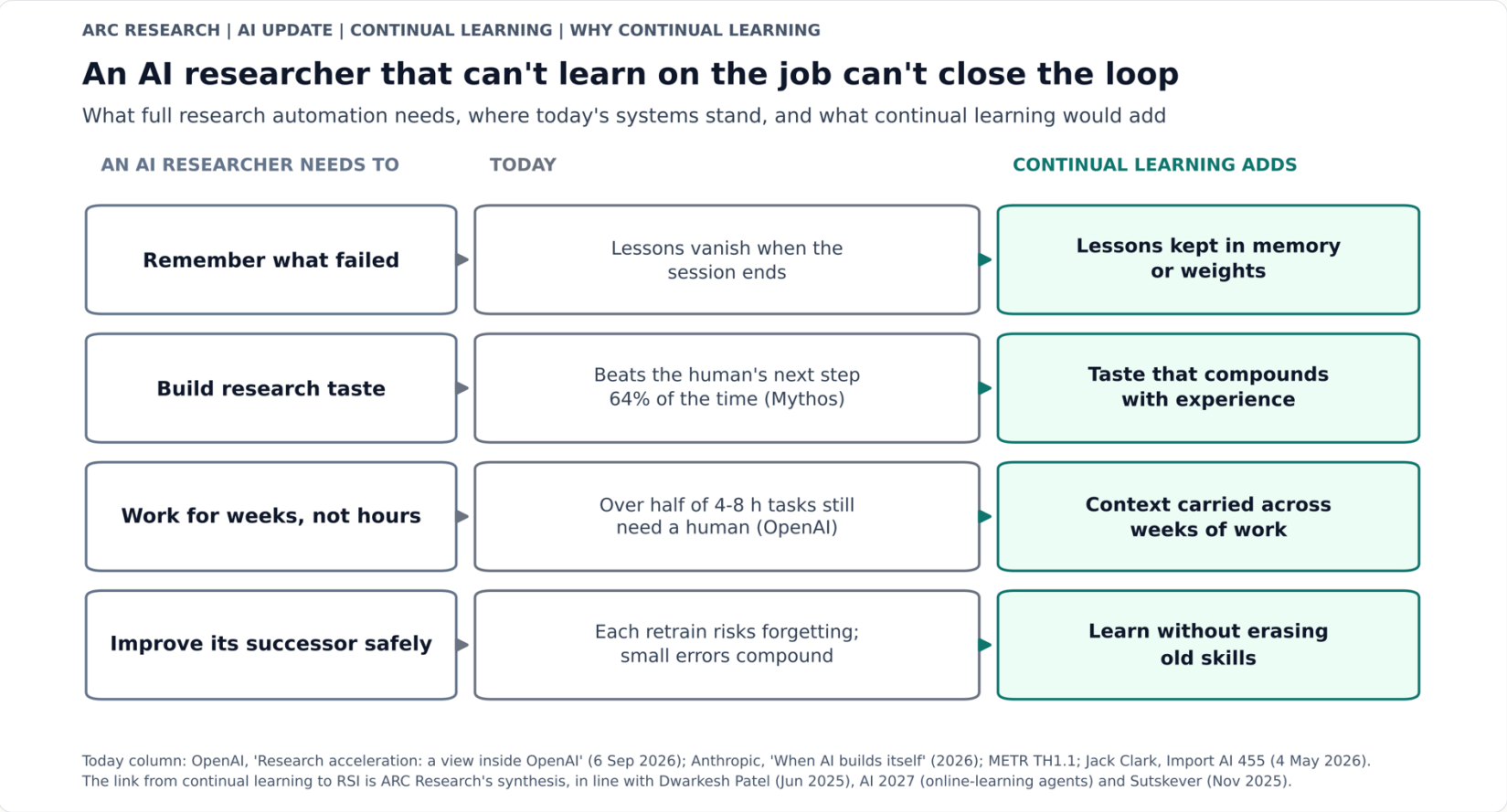
SO WHAT

Training code: 52x vs human 4x

Better next step: 64% of the time

Self-rewriting agent: 20% → 50%

Why RSI needs continual learning



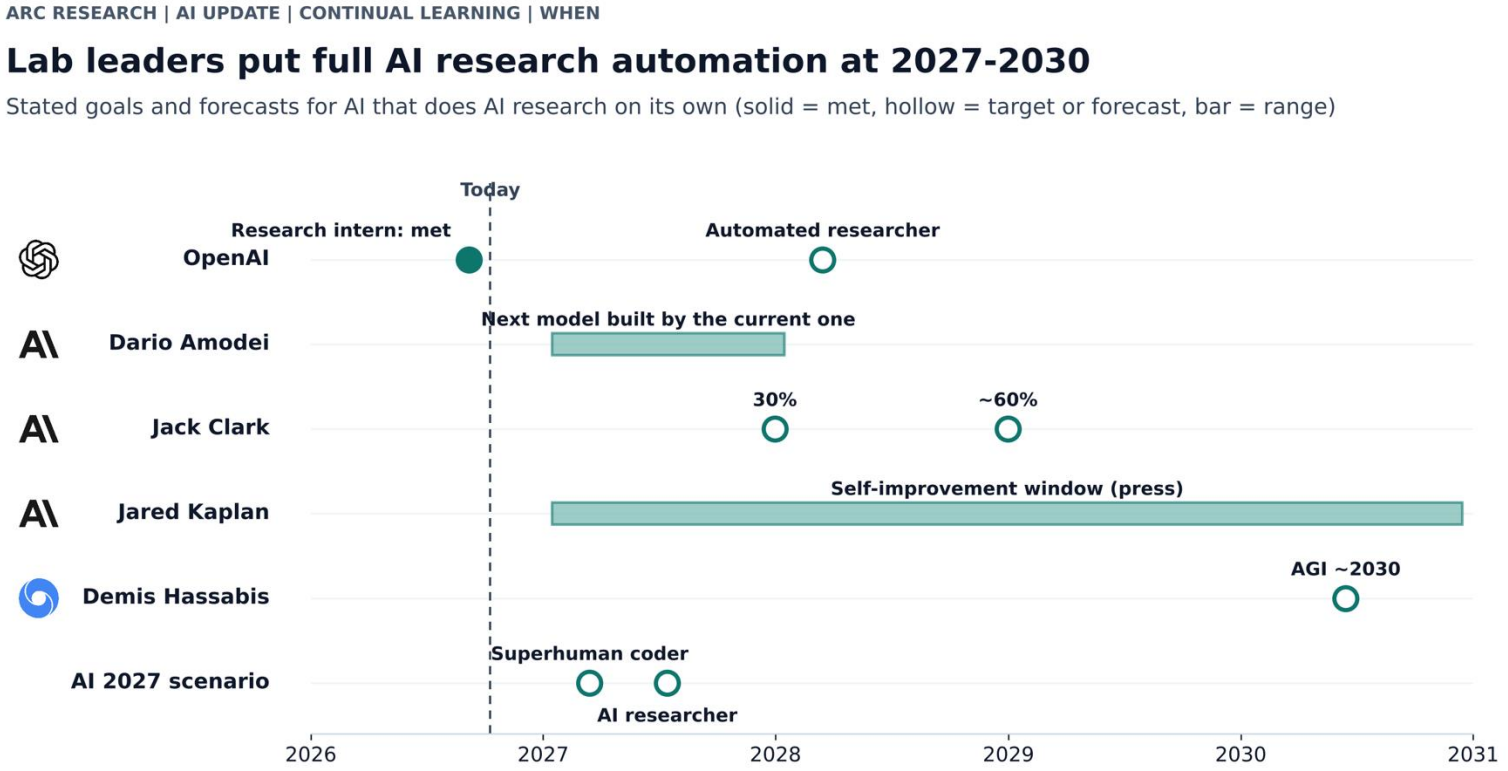
SO WHAT

Today: smart, but forgets

Research needs memory and taste

CL is the missing link

When labs expect AI to do AI research



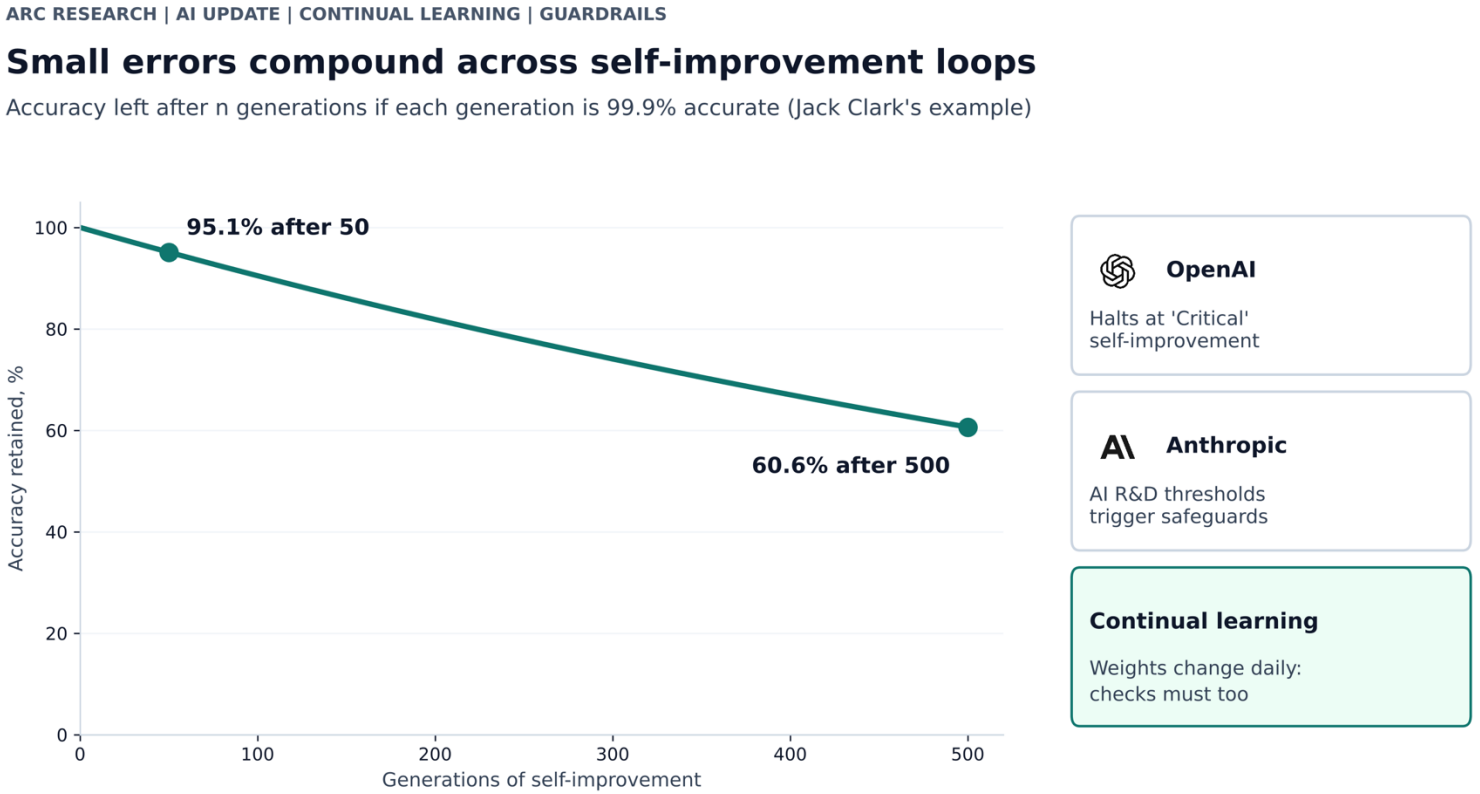
SO WHAT

OpenAI intern: met, Sep 2026

Automated researcher: Mar 2028 target

Clark: ~60% by end-2028

Small errors compound; guardrails matter



Source: Jack Clark, Import AI 455 (4 May 2026): 99.9% per generation gives 95.12% after 50 and 60.5% after 500. Curve = 0.999^n, our calculation (60.6% at 500; Clark rounds to 60.5%).
Guardrails: OpenAI Preparedness Framework v2 (15 Apr 2025), 'AI Self-improvement' Critical level; Anthropic Responsible Scaling Policy AI R&D thresholds (v3.4, 8 Jul 2026).

SO WHAT

99.9% per step → 61% after 500

OpenAI: halt at 'Critical' level

With CL, checks must be continuous

SECTION 6 OF 6

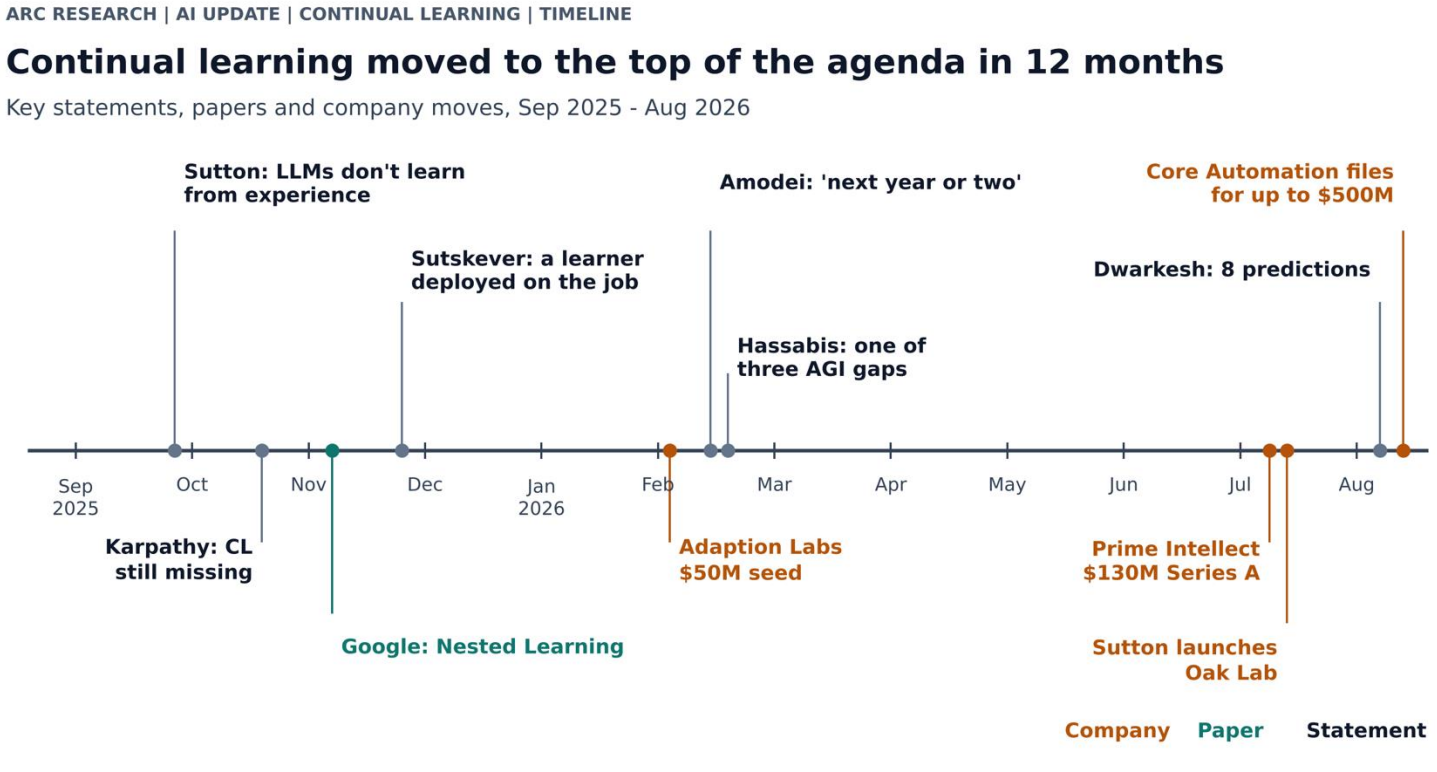
My opinion

Where I think continual learning goes

COMING UP

Twelve months of moves · 40
If it works · 41
My view and the evidence · 42–43
Sources · 44

Twelve months that put CL on the agenda



Sources: Dwarkesh Podcast (Sutton 26 Sep 2025; Karpathy ~17 Oct 2025; Sutskever 25 Nov 2025; Amodei 13 Feb 2026); Google Research blog (7 Nov 2025); Hassabis, New Delhi (reported 18 Feb 2026); Fortune (4 Feb 2026); Prime Intellect blog (8 Jul 2026); The Logic via FourWeekMBA (Jul 2026); dwarkesh.com (7 Aug 2026); Axios (13 Aug 2026).

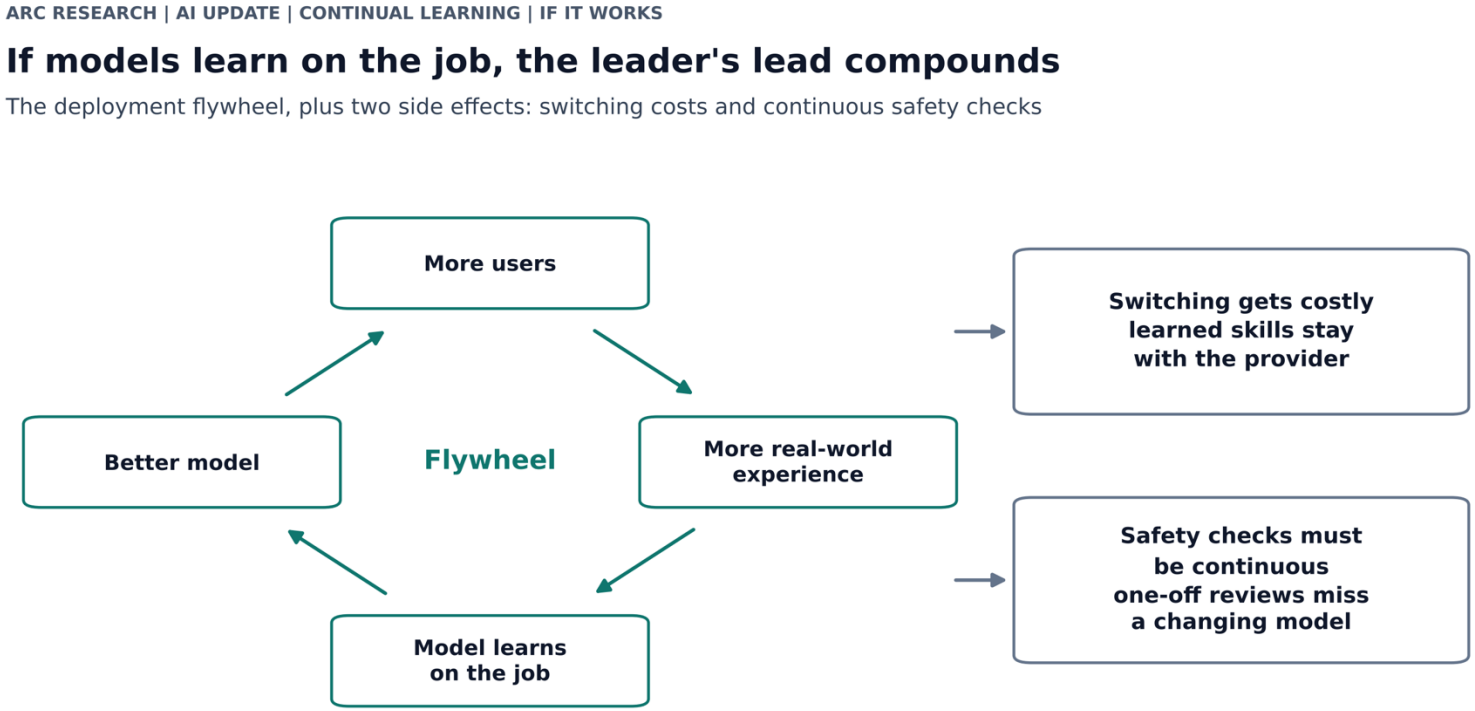
SO WHAT

2025: researchers name the gap

Early 2026: CEOs put dates on it

Mid 2026: capital follows

If models learn on the job, three things change



SO WHAT

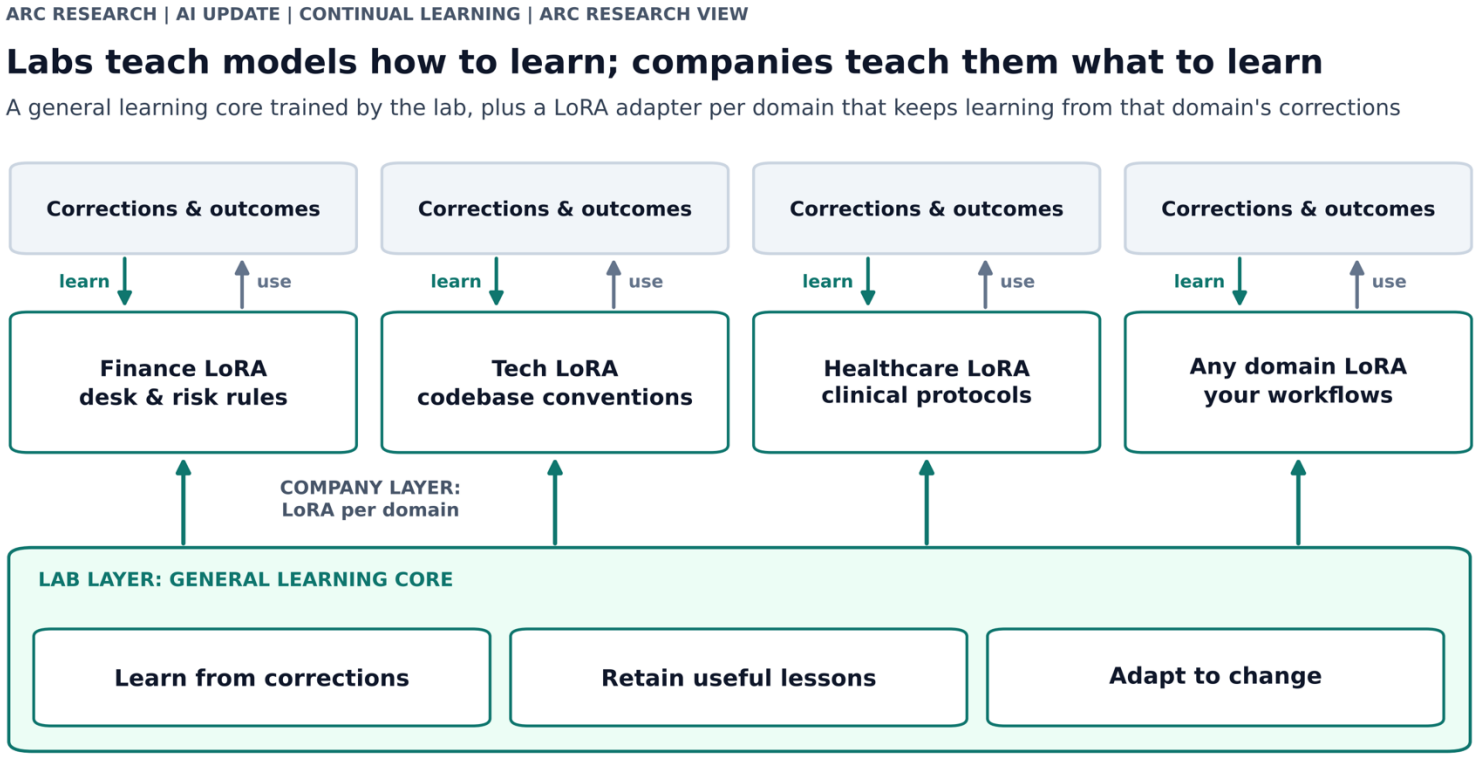
The leader compounds

Switching gets costly

Safety checks must be continuous

MY OPINION (A VIEW, NOT ADVICE)

Labs teach how to learn; companies, what



SO WHAT

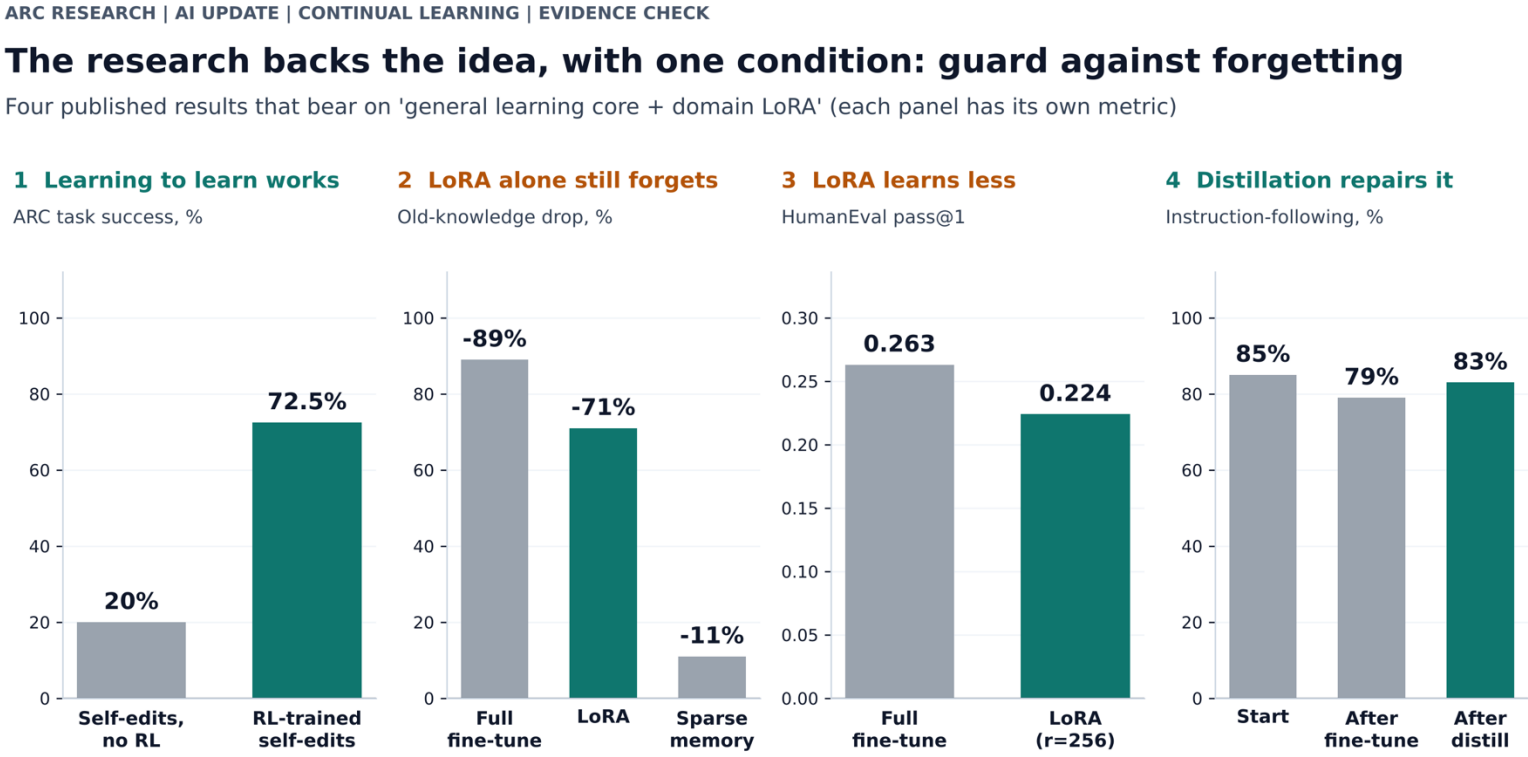
Labs: train the skill of learning

Companies: LoRA per domain

Each domain learns from its own corrections

MY OPINION · EVIDENCE CHECK

It works if LoRA is guarded against forgetting



SO WHAT

Learning to learn: 20% → 72.5%

LoRA alone: 71% forgotten

Fix: distill after each update

Sources

Papers

Lin et al., Sparse Memory Finetuning, arXiv 2510.15103

Behrouz et al., Titans, arXiv 2501.00663

Google Research, Nested Learning (blog, Nov 2025)

Zweiger et al., SEAL, arXiv 2506.10943

Thinking Machines, On-Policy Distillation (Oct 2025)

Statements

Dwarkesh Podcast: Sutton, Karpathy, Sutskever, Amodei

Hassabis remarks (press-reported, Feb and Apr 2026)

Pachocki interview summary (Apr 2026)

Dwarkesh Patel, 8 Predictions (Aug 2026)

Companies

Baseten blog (May 2026); Base Labs agenda (Sep 2026)

Prime Intellect: Lab (Feb 2026), Series A (Jul 2026)

Periodic Labs website; Maginative (Sep 2025)

Fortune on Adaption Labs (Feb 2026)

Axios on Core Automation (Aug 2026)

The Information via Dealroom on Trajectory (Aug 2026)

Pulse 2 on Baseten Series F (Jun 2026)

Sakana AI, Transformer² (Jan 2025); TechCrunch (Nov 2025)

The Information via The Next Web on Thinking Machines (Sep 2026)

FutureSearch evidence review and forecasts (Aug 2026)

RSI

OpenAI, Research acceleration (Sep 2026); Preparedness v2

Anthropic, When AI builds itself (2026); RSP v3.4

METR Time Horizon 1.1; Clark, Import AI 455; DGM